Calibeating Prediction-Powered Inference

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Abstract

We study semisupervised mean estimation with a small labeled sample, a large unlabeled sample, and an available black-box prediction model that may be miscalibrated. This setting arises in randomized trials, biobanks, and other data-rich studies where gold-standard outcomes are expensive but surrogate predictions are readily available. Standard prediction-powered estimators such as augmented inverse-probability weighting (AIPW) (Robins et al., 1994) remain valid under prediction-model misspecification, but they can leave efficiency on the table when the prediction score is poorly calibrated.

We propose a simple post-hoc calibration correction: using the labeled sample, we recalibrate the prediction score and then average the calibrated predictions over the pooled sample. This procedure, which we call *Calibrated Prediction-Powered Inference*, requires no retraining of the original model and can improve efficiency relative to AIPW based on the original uncalibrated score. We study both linear and isotonic calibration. For isotonic calibration, we establish a first-order *calibeating* guarantee: the calibrated estimator can improve on the original score and on simpler post-processing rules, and no further post-processing of the fitted isotonic score yields additional first-order improvement. For linear calibration, we show first-order equivalence to PPI++ and hence the same asymptotic efficiency.

We also clarify how our approach relates to recent prediction-powered inference (PPI) methods. We show that the original PPI estimator of Angelopoulos et al. (2023a) is a special case of AIPW and can be less efficient when the prediction model is accurate, because it does not fully use labeled-sample covariate information. We further show that PPI++ of Angelopoulos et al. (2023b) is AIPW with empirical-efficiency maximization (Rubin and van der Laan, 2008) and is first-order equivalent to linear prognostic-score regression adjustment (Mincer and Zarnowitz, 1969; Hansen, 2008). These results place PPI methods within a broader class of semiparametric debiasing and regression-adjustment methods. In simulations and real-data experiments, our calibrated estimators often improve on PPI and remain competitive with, or outperform, AIPW and PPI++. We provide an accompanying `Python` package, `ppi_aipw`.

1 Introduction

We study semisupervised mean estimation with one labeled dataset $\{(X_i, Y_i)\}_{i=1}^n$ and one unlabeled dataset $\{\widetilde{X}_i\}_{i=1}^N$, where outcomes are observed only for a subset sampled completely at random. This is a classical missing-outcome problem: rich covariate information is available for many individuals, but the outcome of interest is observed only for a smaller subset. Such settings arise when gold-standard outcomes are expensive to collect but prediction scores are readily available, for example in randomized trials with prognostic models trained on historical data (Schuler et al., 2022), biobank analyses using phenotype predictors built from EHR or omics measurements (Hong et al., 2019; Zhang et al., 2019), and computational biology pipelines that supplement slow or costly experimental measurements with model-based predictions (Angelopoulos et al., 2023a; Zrnic and Candès, 2024). The goal is to use the larger unlabeled sample to improve estimation of a

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population mean while accounting for prediction error. This setting has long been studied in missing-data and semiparametric statistics, where methods such as AIPW, TMLE, and related debiased estimators allow flexible machine-learning predictors to be used with valid inference (Robins et al., 1994; van der Laan and Robins, 2003; Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011; Chernozhukov et al., 2018a). More recently, related ideas have reappeared in the machine-learning literature under the name prediction-powered inference (Angelopoulos et al., 2023a,b; Zrnic and Candès, 2024). Closely related questions also arise in covariate adjustment for randomized trials, where machine-learning predictors trained on auxiliary data can improve precision in small samples (Hansen, 2008; Rubin and van der Laan, 2008; Moore and van der Laan, 2009a; Lin, 2013; Schuler et al., 2022; Demirel et al., 2024; Jin, 2026).

A standard approach is to use the labeled sample to debias a plug-in estimator based on a prediction model. If $m(X)$ is a black-box prediction score, the plug-in estimator averages $m(X)$ over the pooled covariate sample, while the labeled sample is used to estimate the residual correction term $\mathbb{E}\{Y - m(X)\}$. For example, the augmented inverse-probability weighted (AIPW) estimator (Robins et al., 1994) averages $m(X)$ over both labeled and unlabeled samples and then adds the average residual over the labeled sample, whereas the PPI estimator (Angelopoulos et al., 2023a) averages $m(X)$ only over the unlabeled covariates before applying the same residual correction. The latter is generally less efficient because it discards information in the labeled covariates. Thus, while these methods protect against bias from imperfect prediction models, they typically treat the prediction score itself as fixed. In practice, however, useful black-box scores are often systematically too high, too low, or too extreme. Such misalignment may arise because the model was trained on a different population, or because of temporal drift, distribution shift, or regularization bias.

This motivates post-hoc calibration of the prediction score. Rather than correcting only the marginal bias $\mathbb{E}\{Y - m(X)\}$, one can use the labeled sample to transform the score so that its numerical values better align with the outcomes they are meant to predict (Zadrozny and Elkan, 2001, 2002; Niculescu-Mizil and Caruana, 2005). An ideal target is a calibrated score $m_n^\star(X)$ satisfying $\mathbb{E}[Y \mid m_n^\star(X)] = m_n^\star(X)$. Under this condition, $\mathbb{E}[m_n^\star(X)] = \mathbb{E}[Y]$, so the corresponding plug-in estimator is automatically unbiased for the population mean. This is attractive in practice because a useful prediction model may already be available, while a relatively small labeled sample can often calibrate its output without retraining the underlying model (Roth, 2022). Classical post-processing methods include Platt scaling (Platt et al., 1999), linear calibration (Mincer and Zarnowitz, 1969), histogram binning (Zadrozny and Elkan, 2001), and isotonic regression (Zadrozny and Elkan, 2002). Such calibration is especially relevant for modern machine-learning predictors, such as neural networks, which are often informative yet systematically miscalibrated (Bella et al., 2010; Guo et al., 2017; Wang, 2023).

Our proposal is therefore simple: fit a black-box regression model, calibrate it on the labeled sample, and then average the calibrated predictions over the pooled covariate sample. This remains a simple "predict, calibrate, and average" procedure, but we show that it also admits an exact debiased interpretation. In particular, the resulting estimator is exactly equal to an AIPW estimator based on the calibrated score, with the residual correction encoded by the calibration step itself. Because calibration can improve the score, it can also improve efficiency relative to AIPW based on the uncalibrated model. Calibration therefore converts a simple plug-in rule into a debiased estimator with attractive efficiency properties.

Contributions. We make three main contributions.

First, in Section 3, we introduce a general *Calibrated Prediction-Powered Inference* framework for semisupervised mean estimation with a small labeled sample and a larger unlabeled one. This framework accommodates familiar post-hoc calibration methods, including linear calibration, Platt scaling, histogram binning, and isotonic regression; see, in particular, Sections 3.2 and 3.3. We focus in particular on two practical estimators, linearly calibrated PPI and isotonic-calibrated PPI, which are simple to implement, tuning-parameter-free, and easy to add to an existing prediction pipeline without retraining the original model. Isotonic calibration is especially appealing because it learns the mean-squared-error-optimal monotone transformation of the prediction score.

Second, we argue that calibration can provide three important statistical benefits. First, it can improve the prediction score and thereby increase the precision of point estimates and confidence intervals. Second,

it makes the calibrated plug-in estimator automatically debiased, because it is exactly equal to the AIPW estimator one would construct from the calibrated score. Third, calibration can produce a fitted score that cannot be improved by further post-processing, in the sense that no transformation of that score within the relevant class of post-processing maps can improve predictive performance or reduce estimator variance to first order. This can be viewed as a form of *calibeating* in the sense of Foster and Hart (2023).

Third, we develop asymptotic theory for linearly calibrated and isotonic-calibrated PPI estimators. For linear calibration, we show that PPI++ is asymptotically equivalent both to linear calibration and to linear regression adjustment using $m(X)$ as a prognostic score. This clarifies the connection between PPI++ and classical covariate-adjustment methods in randomized trials and missing-at-random settings. For isotonic-calibrated PPI, Section 4 establishes asymptotic normality, valid inference, and efficiency results. In particular, we show that isotonic-calibrated PPI achieves the smallest asymptotic variance among estimators based on monotone transformations of the original score, and is therefore at least as efficient as AIPW and PPI using the uncalibrated score. We also show that no further post-processing of the fitted score can improve predictive performance at first order, and that the resulting estimator is first-order equivalent to the oracle estimator based on the conditional mean of the outcome given the calibrated score.

Our calibrated PPI framework can be viewed as a semisupervised specialization of the calibrated debiased machine learning framework developed by van der Laan et al. (2024c). Our estimators can also be interpreted as targeted minimum loss estimators (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011), with calibration playing the role of the targeting step.

Section 2 reviews the two-sample semisupervised model, the AIPW, PPI, and PPI++ estimators, and the main efficiency concepts used throughout the paper. Section 5 presents synthetic and real-data experiments.

Code. The accompanying `Python` package, reproduction code, and documentation are publicly available at github.com/Larsvanderlaan/ppi-aipw. An interactive package website with examples and API documentation is available at larsvanderlaan.github.io/ppi-aipw/. Self-contained code is provided in Appendix M.

1.1 Related work

Although PPI has been presented as a recent development, the underlying problem and many of its estimators are special cases of a broader literature on semiparametric debiasing and regression adjustment, including work on missing data, causal inference, and debiased machine learning.

Semiparametric statistics and missing data. This paper builds on foundational ideas from the missing-data and semiparametric literatures. Rubin (1976) formalized missing at random as a key identification condition for recovering full-data targets from partially observed outcomes. Semiparametric efficiency theory then characterized efficient influence functions, regular asymptotically linear estimators, and one-step debiased procedures (Levit, 1975; Pfanzagl and Wefelmeyer, 1985; Hasminskii and Ibragimov, 1979; Klaassen, 1987; Bickel et al., 1993). In missing-data and causal-inference problems, these ideas led to influence-function-based estimating equations (Robins et al., 1994, 1995; Robins and Rotnitzky, 1995; van der Laan and Robins, 2003; Tsiatis, 2006), often with doubly robust structure (Bang and Robins, 2005). As a result, the efficiency theory for missing-data models under missingness or coarsening at random is now well understood, and the class of regular asymptotically linear estimators in these models can be characterized through their influence functions (Robins et al., 1994, 1995; Robins and Rotnitzky, 1995; van der Laan and Robins, 2003; Tsiatis, 2006).

Debiased machine learning and targeted learning. A large body of work combines the semiparametric framework above with machine-learning-based nuisance estimation, often under the label of debiased machine learning. Flexible learning methods within estimating equations and AIPW-style procedures already appear in van der Laan and Robins (2003). Targeted minimum loss estimation (TMLE), or targeted learning, updates an initial nuisance estimator in a parameter-targeted way so that the resulting plug-in estimator is debiased; in the present setting, the main nuisance is the outcome regression (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011; Gruber and Van Der Laan, 2009; Hoffman, 2020; Ross et al., 2025). These approaches are often combined with sample splitting and cross-fitting to accommodate generic machine-learning estimators (Schick, 1986; Klaassen, 1987; Zheng and Van Der Laan, 2010; van der Laan and Rose, 2011; Chernozhukov et al.,

2018a). Double machine learning emphasizes similar ingredients—orthogonality, cross-fitting, and influence-function expansions—for inference with modern nuisance estimators (Chernozhukov et al., 2018a, 2022), and can be understood within the same semiparametric framework (van der Laan, 2019; Díaz, 2020; Chen et al., 2026). For sufficiently stable estimators, cross-fitting can itself be relaxed (Chen et al., 2022). Closely related work studies balancing-weight estimators (Hainmueller, 2012; Imai and Ratkovic, 2014; Zubizarreta, 2015; Lendle et al., 2015; Chattopadhyay et al., 2020; Hejazi and van der Laan, 2023), which often admit equivalent regression-adjustment or augmented representations (Chattopadhyay and Zubizarreta, 2023; Bruns-Smith et al., 2025; Rotnitzky et al., 2025). In completely missing-at-random settings, including semisupervised learning and randomized trials, these approaches allow flexible black-box nuisance estimation while still supporting valid inference under appropriate conditions (Moore and van der Laan, 2009a,b; Rosenblum and van der Laan, 2009; Højbjerre-Frandsen et al., 2025; Højbjerre-Frandsen and Schuler, 2026).

Semisupervised inference. Related work studies semisupervised or surrogate-assisted estimation in settings where the primary outcome is observed only on a limited subset, while auxiliary outcomes or surrogate measurements are available more broadly (Pepe, 1992; Pepe et al., 1994; Chen, 2000; Chen et al., 2008; Cheng et al., 2021; Ji et al., 2025; Kallus and Mao, 2025). Two-sample semisupervised inference is a special case of two-stage sampling designs (Scott and Holt, 1982; Rose and van der Laan, 2011; Hejazi et al., 2021; Qiu et al., 2026) and, more generally, of data fusion, for which a substantial semiparametric efficiency literature has been developed (Li and Luedtke, 2023; Li et al., 2025; Graham et al., 2024; Xu et al., 2025). More recent papers have used the term *prediction-powered inference* (PPI) for the semisupervised mean-estimation setting in which a black-box predictor is combined with a small labeled sample and a large unlabeled sample (Angelopoulos et al., 2023a,b; Zrnic and Candès, 2024; Xu et al., 2025; Song et al., 2026; Poulet et al., 2025; Lee and Kim, 2026). These papers also helped popularize such methods for a broader machine-learning audience and provided accessible software. Closely related questions have long been studied in randomized trials, where prognostic scores or other regression adjustments learned from larger historical or auxiliary data sources are used to improve precision in smaller trials (Hansen, 2008; Rosenblum and van der Laan, 2009; Moore and van der Laan, 2009a,b; Moore et al., 2011; Schuler et al., 2022; Balzer et al., 2024; Højbjerre-Frandsen et al., 2025). Methodologically, however, PPI estimators are best understood as instances or variants of existing semiparametric procedures. For example, PPI++ is an AIPW estimator (Robins et al., 1994; van der Laan and Robins, 2003) obtained by empirical efficiency maximization (Rubin and van der Laan, 2008) over a scaling parameter, while cross-PPI (Zrnic and Candès, 2024) is simply AIPW with cross-fitting. More fundamentally, the ability to combine black-box prediction models with valid inference is a consequence of the double robustness of AIPW when the missingness probability is known, and is not specific to semisupervised mean estimation (Bang and Robins, 2005; Seaman and Vansteelandt, 2018; Rotnitzky et al., 2021).

From missing data to causal inference. From the perspective of estimation and efficiency theory, semisupervised mean estimation with missing-at-random outcomes is equivalent to randomized experiments, controlled trials, and survey-sampling designs once the missingness indicator is reinterpreted as a treatment or sampling indicator, so that the relevant potential outcomes are viewed as missing (Rubin, 1978; van der Laan and Robins, 2003; Ding and Li, 2018; Mozer, 2026). This equivalence allows tools from covariate adjustment, causal inference, and model-assisted estimation to be transferred directly to the semisupervised setting; see, for example, Mozer (2026) for results relating prediction-powered inference to model-assisted survey-sampling estimators. The two-sample formulation we study is also equivalent to a one-sample i.i.d. model (Li and Luedtke, 2023). In particular, our methods apply to inference on the counterfactual mean $\mathbb{E}[Y(1)]$ in a randomized trial with binary treatment, where treated units form the labeled sample and control units contribute covariate information only; see Appendix D.

Calibration. A final relevant thread concerns calibration. Classical post-processing methods such as Platt scaling (Platt et al., 1999; Cox, 1958; Gupta and Ramdas, 2023), linear calibration (Mincer and Zarnowitz, 1969; Chernozhukov et al., 2018b; Leng and Dimmery, 2021), isotonic regression (Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005), and histogram binning (Zadrozny and Elkan, 2001; Gupta and Ramdas, 2021) improve predictive reliability without retraining the underlying model. The idea has deeper roots in the forecasting literature (Mincer and Zarnowitz, 1969; Lichtenstein et al., 1977; Vovk, 1992; Vovk et al., 2005; Gneiting et al., 2007; Lambert, 2011); see also the historical discussion in Lee et al. (2023). More recent work studies calibration under broader losses and prediction tasks (Jung et al., 2021; Noarov and Roth, 2023;

van der Laan et al., 2023b; Whitehouse et al., 2024; van der Laan and Alaa, 2025), as well as applications to conformal prediction (van der Laan and Alaa, 2024, 2025), importance-weight stabilization (Gutman et al., 2022; Deshpande and Kuleshov, 2023; Ballinari and Bearth, 2024; van der Laan et al., 2024b), treatment effect estimation (Chernozhukov et al., 2018b; van der Laan et al., 2023b; Whitehouse et al., 2024), and value prediction in offline reinforcement learning (van der Laan and Kallus, 2025). Distribution-free guarantees for histogram binning are developed by Gupta and Ramdas (2021), while van der Laan et al. (2023b); van der Laan and Alaa (2025) establish asymptotic distribution-free guarantees for isotonic calibration. Most closely related to our work is recent research connecting calibration to debiased estimation of statistical functionals (van der Laan et al., 2024c, 2023a, 2025b). In particular, van der Laan et al. (2024c) studies calibrated debiased machine learning for doubly robust inference, which in our setting reduces to calibrating the outcome regression and exploiting the resulting plug-in representation, while van der Laan et al. (2025b) studies calibrated plug-in regression estimators in reinforcement learning. More broadly, van der Laan et al. (2023a) shows how plug-in calibration can support regression and treatment effect estimation, including superefficient inference through dimension reduction.

2 Two-Sample Semisupervised Setup

2.1 Data structure and notation

We observe two independent datasets,

$$\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n \qquad \text{and} \qquad \mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N.$$

The labeled sample consists of i.i.d. draws from a distribution P_0 on $\mathcal{X} \times \mathbb{R}$, while the unlabeled sample consists of i.i.d. draws from the distribution $P_{0,X}$ of X under P_0. Our target is the population mean

$$\psi_0 := \mathbb{E}_{P_0}[Y], \qquad \mu_0(x) := \mathbb{E}_{P_0}[Y \mid X = x].$$

This can also be written as

$$\psi_0 = \int y \, dP_0(x, y) = \int \mu_0(x) \, dP_{0,X}(x).$$

For functions $a : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ and $b : \mathcal{X} \to \mathbb{R}$, define the empirical means over the labeled and unlabeled samples by

$$\mathbb{P}_n^L a := \frac{1}{n} \sum_{i=1}^n a(X_i, Y_i), \qquad \mathbb{P}_N^U b := \frac{1}{N} \sum_{j=1}^N b(\widetilde{X}_j).$$

For readability, we will sometimes write $\mathbb{P}_n^L\{a(X, Y)\}$ instead of $\mathbb{P}_n^L a$. When a function depends only on X, we also write

$$\mathbb{P}_n^L\{b(X)\} := \frac{1}{n} \sum_{i=1}^n b(X_i).$$

Throughout, let $M := n + N$ and $\rho_n := n/M$ denote the labeled fraction. Then

$$\frac{1}{n+N} \left\{ \sum_{i=1}^n f(X_i) + \sum_{j=1}^N f(\widetilde{X}_j) \right\} = \rho_n \mathbb{P}_n^L f + (1 - \rho_n) \mathbb{P}_N^U f.$$

In other words, the empirical mean of $f(X)$ over the pooled covariate sample is a weighted average of the empirical means from the labeled and unlabeled samples.

Assumption 1 (Two-sample design)**.** The following hold:

(i) $\mathcal{D}_L$ is i.i.d. from P_0, and $\mathcal{D}_U$ is i.i.d. from $P_{0,X}$.

(ii) $\mathcal{D}_L$ and $\mathcal{D}_U$ are independent.

(iii) Y has finite second moment under P_0.

(iv) With $M := n + N$, the labeled fraction $\rho_n = n/M$ satisfies $\rho_n \to \rho_0$ for some $\rho_0 \in (0, 1)$.

Remark 1. The labeled sample carries the outcome information, while the unlabeled sample sharpens estimation of the marginal covariate distribution. The pooled i.i.d. missing-data formulation in Appendix K is a special case of this two-sample setup.

2.2 Review of AIPW, PPI, and semiparametric efficiency

The AIPW class. We begin with the augmented inverse-probability weighted (AIPW) estimator in the unrestricted two-sample model (Robins et al., 1994). It is helpful to view AIPW not as a single estimator, but as a family of estimators indexed by a score function $f : \mathcal{X} \to \mathbb{R}$ with finite second moment. For any such f, define

$$\widehat{\psi}(f) := \rho_n \mathbb{P}_n^L\{f(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{f(\widetilde{X})\} + \mathbb{P}_n^L\{Y - f(X)\}. \tag{1}$$

This estimator combines a plug-in term that imputes Y from $f(X)$ with a residual correction from the labeled sample. For any fixed f, $\widehat{\psi}(f)$ is unbiased in finite samples for $\psi_0 = \mathbb{E}_{P_0}[Y]$. In particular, $f(X) = 0$ yields the empirical mean estimator $\mathbb{P}_n^L\{Y\}$. Thus, the choice of f affects efficiency, through the estimator's variance, but not validity, in the sense that the resulting confidence intervals remain valid. This makes AIPW a "safe" estimator (Xu et al., 2025). This property can be viewed as a particularly simple instance of double robustness[1] (Bang and Robins, 2005). The same basic principle underlies the use of flexible machine learning for covariate adjustment in randomized trials while retaining valid inference (Moore and van der Laan, 2009a,b; Schuler et al., 2022; Højbjerre-Frandsen et al., 2025).

Asymptotic linearity and efficiency. To characterize the leading large-sample error in the two-sample model, we decompose the estimation error into one contribution from the labeled sample and one from the unlabeled sample. We say that $\widehat{\psi}$ is asymptotically linear with influence pair (D^L, D^U) if

$$\widehat{\psi} - \psi_0 = \frac{1}{M}\sum_{i=1}^{n} D^L(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^{N} D^U(\widetilde{X}_j) + o_p(M^{-1/2}),$$

where $\mathbb{E}_{P_0}[D^L(X, Y)] = 0$ and $\mathbb{E}_{P_{0,X}}[D^U(\widetilde{X})] = 0$. By the central limit theorem and the independence of the two samples,

$$\sqrt{M}\,(\widehat{\psi} - \psi_0) \rightsquigarrow N\Big(0,\ \rho_0 \mathrm{Var}_{P_0}\{D^L(X, Y)\} + (1 - \rho_0)\mathrm{Var}_{P_{0,X}}\{D^U(\widetilde{X})\}\Big).$$

The influence pair therefore determines the estimator's asymptotic variance and provides a convenient way to compare efficiency across different choices of f. The next result uses this representation to characterize the AIPW class, identify the efficient influence pair in the unrestricted two-sample model, and thereby obtain the semiparametrically efficient variance bound (Bickel et al., 1993).

Proposition 1 (Influence pair for the AIPW class)**.** *Under Assumption 1, for any fixed $f : \mathcal{X} \to \mathbb{R}$ with finite second moment, the estimator $\widehat{\psi}(f)$ is asymptotically linear with influence pair (D_f^L, D_f^U), where*

$$D_f^L(X, Y) = \tilde{f}(X) - \psi_0 + \rho_0^{-1}\{Y - \tilde{f}(X)\}, \tag{2}$$

$$D_f^U(\widetilde{X}) = \tilde{f}(\widetilde{X}) - \psi_0, \tag{3}$$

[1] In more general missing-at-random settings with unknown, covariate-dependent missingness probabilities, the analogous AIPW estimator remains doubly robust asymptotically, but it no longer enjoys the same finite-sample unbiasedness as in the present known-ρ_n setting (Seaman and Vansteelandt, 2018).

with $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$ *denoting the recentered version of* $f(X)$*. The resulting asymptotic variance of* $\sqrt{M}\,(\widehat{\psi}(f) - \psi_0)$ *is*

$$\mathrm{Var}\{\mu_0(X)\} + \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big] + \frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

This variance is minimized over the AIPW class when $\tilde{f} = \mu_0$*, in particular at* $f = \mu_0$*. Therefore, the efficient influence pair for* $\psi_0 = \mathbb{E}_{P_0}[Y]$ *in the unrestricted two-sample model is* $(D^L_{\mu_0}, D^U_{\mu_0})$*, and the corresponding asymptotic variance is the efficiency bound*[2]*.*

A key insight of Proposition 1 is that the limiting variance of the AIPW estimator depends on the score function f only through the mean squared error of its recentered version $\tilde{f}$, namely $\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big]$. Thus, within the AIPW class, efficiency is maximized by choosing f so that $\tilde{f}$ approximates $\mu_0(X)$ as well as possible, with equality when $\tilde{f} = \mu_0$, in particular at $f = \mu_0$. This shows that full efficiency generally requires using the full covariate information X, or at least a summary rich enough to recover $\mu_0(X)$ (Robins and Rotnitzky, 1995). In practice, this motivates taking f to be an estimated prediction score $m(X)$, learned using flexible black-box methods.

AIPW, PPI, and empirical-efficiency maximization. We now record several familiar choices of f within the class (1). The most common choice is $f(X) = m(X)$, where $m(X)$ is a prediction score, yielding the usual augmented inverse-probability weighted (AIPW) estimator (Robins et al., 1994):

$$\widehat{\psi}_{\mathrm{AIPW}} = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

Equivalently,

$$\widehat{\psi}_{\mathrm{AIPW}} = \frac{1}{M}\left\{\sum_{i=1}^n m(X_i) + \sum_{j=1}^N m(\widetilde{X}_j) + \frac{1}{\rho_n}\sum_{i=1}^n \{Y_i - m(X_i)\}\right\},$$

where ρ_n^{-1} is the inverse probability of observing the outcome.

Another estimator in this class was considered by Angelopoulos et al. (2023a) under the name PPI,[3] and we adopt that terminology here. It corresponds to the score choice

$$f(X) = \frac{m(X)}{1-\rho_n}.$$

Substituting this choice into (1) gives

$$\widehat{\psi}_{\mathrm{PPI}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

Relative to $\widehat{\psi}_{\mathrm{AIPW}}$, $\widehat{\psi}_{\mathrm{PPI}}$ uses only the unlabeled sample in the plug-in term and rescales the score by the factor $(1-\rho_n)^{-1} > 1$. By Proposition 1, this helps only if the rescaled score is closer to $\mu_0(X)$ in mean squared error. Thus, PPI is inefficient when $m = \mu_0$, and is typically less efficient when m is already reasonably accurate. When ρ_n is small and m is sufficiently inaccurate, however, neither estimator is necessarily more efficient than the other (Han, 2012; Rotnitzky et al., 2012).

Because both $\widehat{\psi}_{\mathrm{AIPW}}$ and $\widehat{\psi}_{\mathrm{PPI}}$ use scores in the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, it is natural to choose the scaling factor data-adaptively rather than fixing it in advance. One way to do so is to minimize the estimator's empirical influence-function variance over this class:

$$\sigma_n^2(f) := \rho_n \mathbb{P}_n^L\left[\left\{f(X) - \widehat{\psi}(f) + \rho_n^{-1}(Y - f(X))\right\}^2\right] + (1-\rho_n)\mathbb{P}_N^U\left[\left\{f(\widetilde{X}) - \widehat{\psi}(f)\right\}^2\right].$$

[2]Appendix A shows that every regular estimator is asymptotically equivalent to $\widehat{\psi}(f)$ for some score function f. Thus, the AIPW family characterizes the full class of regular estimators and the semiparametric efficiency bound in this model.

[3]The PPI and PPI++ estimators are algebraically equivalent to the model-assisted survey-design estimators of Cassel et al. (1976) and Särndal et al. (2003), respectively; see Mozer (2026) for a review.

That is, one chooses λ to minimize $\sigma_n^2(f)$ over $f(X) = \lambda m(X)$, a procedure known as empirical-efficiency maximization (Rubin and van der Laan, 2008; van der Laan and Robins, 2003). The resulting estimator, called PPI++ by Angelopoulos et al. (2023b), is simply the AIPW estimator with an empirically rescaled prediction rule $\hat{\lambda} m(X)$, where $\hat{\lambda}$ estimates the population-optimal scaling coefficient

$$\lambda^\star = \frac{\mathrm{Cov}\big(Y, m(X)\big)}{\mathrm{Var}\big(m(X)\big)}. \tag{4}$$

The same principle extends beyond simple rescaling. For example, one can minimize $\sigma_n^2(f)$ over richer classes such as $\{\lambda m(X) + \beta_{\text{lin}}^\top X : \lambda \in \mathbb{R},\ \beta_{\text{lin}} \in \mathbb{R}^d\}$, over monotone transformations of the score, or over a reproducing kernel Hilbert space (Appendix C). More generally, the same criterion can be used in cross-validation to choose among candidate estimator classes (Rubin and van der Laan, 2008).

3 Calibrated Prediction-Powered Inference

Section 2.2 showed that AIPW provides a safe debiasing template, while the choice of score $f(X)$ determines efficiency. We now introduce the *calibrated prediction-powered inference* framework, which improves an available prediction score $m(X)$ through post-processing rather than retraining. Using the labeled sample, we learn a transformation of $m(X)$ within a chosen class and then use the resulting calibrated score to construct an estimator of the target mean. As we will show, this estimator admits both a simple plug-in form and an exact AIPW representation. Different choices of calibration class lead to different calibrated PPI estimators, including the linear and isotonic procedures studied below.

3.1 General estimator and exact AIPW representation

Let $m : \mathcal{X} \to \mathbb{R}$ denote a black-box regression model. A calibration procedure uses the labeled scores $m(X_1), \ldots, m(X_n)$ and outcomes $Y_1, \ldots, Y_n$ to learn a post-hoc map f_n, and then defines the calibrated regression model

$$m_n^\star := f_n \circ m.$$

The resulting predictions on the labeled and unlabeled samples are $m_n^\star(X_i)$ for $i = 1, \ldots, n$ and $m_n^\star(\widetilde{X}_j)$ for $j = 1, \ldots, N$. Intuitively, the map f_n updates the original score $m(X)$ so that its numerical values better align with the outcome.

This post-processing step has two main goals. First, it aims to improve the original prediction score m through a simple transformation, without retraining the underlying model, ideally by improving calibration without sacrificing predictive performance. Second, it is designed so that the resulting plug-in estimator is automatically unbiased, without requiring an explicit residual correction.

Given the post-processed score $m_n^\star$, define the calibrated PPI estimator by

$$\widehat{\psi}_{\text{cal}} := \frac{1}{n+N} \left\{ \sum_{i=1}^{n} m_n^\star(X_i) + \sum_{j=1}^{N} m_n^\star(\widetilde{X}_j) \right\}. \tag{5}$$

Equivalently,

$$\widehat{\psi}_{\text{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\}.$$

To distinguish the class used to learn the post-processing map from the class used to assess the resulting prediction score, we introduce two function classes. Let $\mathcal{H}$ be an optimization class of real-valued maps $h : \mathbb{R} \to \mathbb{R}$, and let $\mathcal{F}$ be a linear calibration class of real-valued maps $f : \mathbb{R} \to \mathbb{R}$ containing the constant functions and the identity map. We learn the post-processing map f_n by empirical risk minimization over $\mathcal{H}$, that is,

$$f_n \in \arg\min_{h \in \mathcal{H}} \mathbb{P}_n^L\big[(Y - h(m(X)))^2\big]. \tag{6}$$

We then ask what calibration property the resulting post-processed score $m_n^\star := f_n \circ m$ satisfies with respect to the class $\mathcal{F}$.

We say that $m_n^\star$ is *empirically calibrated over* $\mathcal{F}$ if, for every $f \in \mathcal{F}$,

$$\mathbb{P}_n^L\big[(Y - m_n^\star(X))^2\big] \le \mathbb{P}_n^L\big[(Y - f(m_n^\star(X)))^2\big], \tag{7}$$

where $\mathbb{P}_n^L$ denotes the empirical mean over the labeled sample. In words, once $m_n^\star$ has been formed, no further transformation of $m_n^\star$ within $\mathcal{F}$ can improve the empirical squared-error fit.

Because $\mathcal{F}$ is linear, this condition is equivalent to the first-order optimality conditions

$$\mathbb{P}_n^L\big[f(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0, \qquad f \in \mathcal{F}. \tag{8}$$

Since $\mathcal{F}$ contains the constant functions and the identity map, these conditions imply, in particular,

$$\mathbb{P}_n^L\{Y - m_n^\star(X)\} = 0, \qquad \mathbb{P}_n^L\big[m_n^\star(X)\{Y - m_n^\star(X)\}\big] = 0.$$

The first identity is an empirical mean-calibration condition: the residuals of the calibrated score average to zero on the labeled sample, yielding the exact AIPW representation below. The second shows that $m_n^\star$ is already optimal over the one-dimensional scaling class $\{\lambda m_n^\star : \lambda \in \mathbb{R}\}$, so no rescaling of $m_n^\star$ can further reduce empirical squared error.

The next result gives the basic structural property of calibrated PPI estimators.

Theorem 1 (Exact AIPW representation under empirical calibration)**.** *Suppose the calibrated predictions satisfy equation (8), where $\mathcal{F}$ contains all affine maps. Then, for any $\widehat{w} \in \mathcal{F}$,*

$$\widehat{\psi}_{\text{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \mathbb{P}_n^L\big[\widehat{w}(m_n^\star(X))\{Y - m_n^\star(X)\}\big]. \tag{9}$$

In particular, taking $\widehat{w} \equiv 1$ yields the exact AIPW representation

$$\widehat{\psi}_{\text{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m_n^\star(X)\}.$$

If, in addition, $\widehat{w}_n^\star \in \mathcal{F}$ satisfies

$$\mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))\, f(m_n^\star(X))\big] = \rho_n \mathbb{P}_n^L\{f(m_n^\star(X))\} + (1-\rho_n)\mathbb{P}_N^U\{f(m_n^\star(\widetilde{X}))\}, \qquad f \in \mathcal{F}, \tag{10}$$

then

$$\widehat{\psi}_{\text{cal}} = \mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))Y\big]. \tag{11}$$

Theorem 1 shows that empirical calibration does more than improve prediction: it gives the calibrated PPI estimator an exact debiased representation. Specifically, $\widehat{\psi}_{\text{cal}}$ is exactly an AIPW estimator with score $m_n^\star$. The same calibration conditions also yield an equivalent balancing-weight representation, under which $\widehat{\psi}_{\text{cal}}$ is a weighted average of Y over the labeled sample, with the unlabeled sample entering only through the weights. This representation makes clear how unlabeled information is incorporated and enables familiar weighting diagnostics, such as balance checks and inspection of the weight distribution.

Full versus linear calibration. We focus on two main cases. In the nonparametric case, $\mathcal{F}$ is the class of all transformations $f : \mathbb{R} \to \mathbb{R}$, which we call *full calibration*. At the population level, a predictor is fully calibrated if no further transformation can improve its mean squared prediction error. In practice, however, the fitted post-processing map f_n is obtained by minimizing empirical squared error over a chosen optimization class $\mathcal{H}$, which need not coincide with $\mathcal{F}$. The class $\mathcal{H}$ acts as a regularizer: without such a restriction, one could overfit the labeled sample, achieving perfect empirical fit but poor prediction and poor population-level calibration. For example, $\mathcal{H}$ may consist of piecewise constant or monotone functions, as in histogram regression or isotonic regression (Section 3.2). In the parametric case, both $\mathcal{H}$ and $\mathcal{F}$ are restricted to affine transformations $t \mapsto a + bt$, which we call *linear calibration*. This corresponds to fitting the working linear model $a + b\,m(X)$ for Y (Section 3.3).

Why calibration can improve efficiency. The representation in Theorem 1 explains why calibration yields a valid debiased estimator; the remaining question is why it may also improve precision. To see this, suppose that the fitted post-processing map f_n converges to a population minimizer $g_0 \in \mathcal{H}$, and hence that the post-processed score $m_n^\star = f_n \circ m$ converges to

$$m_0(X) = g_0\{m(X)\}, \qquad g_0 \in \arg\min_{g \in \mathcal{H}} \mathbb{E}_{P_0}\left[\{Y - g(m(X))\}^2\right].$$

Thus, m_0 is the best predictor obtainable from the original score m using post-processing maps in the class $\mathcal{H}$. When $m_n^\star$ is close to m_0, $\widehat{\psi}_{\text{cal}}$ behaves like the AIPW estimator that uses m_0. Proposition 1 therefore implies that this limiting estimator has asymptotic variance no larger than that of the AIPW estimator based on any $\theta \circ m$ with $\theta \in \mathcal{H}$. In particular, if $\mathcal{H}$ contains the identity map, as it does for affine or monotone transformations, then m itself is among the admissible post-processings. It follows that the estimator based on m_0 is asymptotically at least as efficient as the AIPW estimator based on the original score m. Calibration can therefore improve efficiency insofar as m_0 provides a better regression adjustment than m.

3.2 Isotonic-calibrated PPI

We now describe two simple second-stage calibration methods that fit naturally into the $(\mathcal{H}, \mathcal{F})$ framework: histogram binning and isotonic regression (Zadrozny and Elkan, 2001, 2002; Niculescu-Mizil and Caruana, 2005). Both are fit using the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^n$, but they use different optimization classes $\mathcal{H}$.

For histogram binning, $\mathcal{H}$ is the class of piecewise-constant maps on a prespecified partition of the score space (Gupta et al., 2020; Gupta and Ramdas, 2021). One may take $\mathcal{F} = \mathcal{H}$. In fact, because the fitted score is constant within each bin, any further transformation of that score is also piecewise constant on the same partition. Thus histogram binning is empirically calibrated not only over $\mathcal{H}$, but also fully calibrated. In this sense, a post-processing map learned over a restricted class $\mathcal{H}$ can still yield full empirical calibration.

For isotonic calibration, $\mathcal{H} = \mathcal{F}_{\text{iso}}$, where $\mathcal{F}_{\text{iso}}$ denotes the class of nondecreasing real-valued functions. The calibrator is fit by isotonic regression via the pooled-adjacent-violators algorithm (Barlow and Brunk, 1972). The fitted score is piecewise constant on a data-adaptive partition of the score space determined by the pooled blocks. As with histogram binning, any further transformation remains constant on the same blocks, so isotonic calibration also yields full empirical calibration. We focus on isotonic regression because it does not require prespecified bins: the partition is learned adaptively from the data, and the method is lightweight, tuning-free, and easy to implement with widely available software.

The isotonic-calibrated PPI estimator is

$$\widehat{\psi}_{\text{iso}} := \frac{1}{n+N}\left\{\sum_{i=1}^n m_{n,\text{iso}}^\star(X_i) + \sum_{j=1}^N m_{n,\text{iso}}^\star(\widetilde{X}_j)\right\},$$

where

$$m_{n,\text{iso}}^\star(X) = \hat{f}\{m(X)\}, \qquad \hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \big\{Y_i - f(m(X_i))\big\}^2.$$

The key calibration property of isotonic regression, and likewise of histogram binning, is the following.

Proposition 2 (Isotonic calibration is fully calibrated)**.** *For every function $h : \mathbb{R} \to \mathbb{R}$,*

$$\mathbb{P}_n^L\left[h(m_{n,\text{iso}}^\star(X))\{Y - m_{n,\text{iso}}^\star(X)\}\right] = 0. \tag{12}$$

Moreover,

$$\mathbb{P}_n^L\left[(Y - m_{n,\text{iso}}^\star(X))^2\right] \le \mathbb{P}_n^L\left[(Y - m(X))^2\right].$$

Consequently, the representations in Theorem 1 hold with $\mathcal{F}$ taken to be the class of all real-valued functions.

Algorithm 1 Isotonic-calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $\hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \{Y_i - f(T_i)\}^2$
3: Set $m^\star_{n,\text{iso}}(X_i) = \hat{f}(T_i)$ and $m^\star_{n,\text{iso}}(\widetilde{X}_j) = \hat{f}(\widetilde{T}_j)$
4: Form $\widehat{\psi}_{\text{iso}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\text{iso}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{iso}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\text{iso}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}$
6: Set $\widehat{\text{SE}}^2_{\text{iso}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{iso}}$

Remark 2 (Implementation)**.** Isotonic regression is widely available in standard software. A standard algorithm is the pooled-adjacent-violators algorithm (PAVA) (Barlow and Brunk, 1972). Although it is generally fast, it may become slow at ultra-large sample sizes. A scalable alternative is to use regression tree software with monotonicity constraints, such as XGBoost or LightGBM (see Appendix M). An advantage of tree-based software is that the fit can be regularized by constraining or tuning the number of leaves, the maximum tree depth, or the minimum number of observations per leaf.

3.3 Linearly calibrated PPI

For small labeled samples, one may prefer smoother parametric calibration rules to histogram binning or isotonic regression. We focus on linear calibration, in which $\mathcal{H} = \mathcal{F} = \mathcal{F}_{\text{lin}}$, where $\mathcal{F}_{\text{lin}}$ denotes the class of affine maps $t \mapsto at + b$. Thus, the post-processing step is learned over the same class with respect to which calibration is assessed. Compared with isotonic regression, this imposes a more restrictive parametric form, akin in spirit to Platt scaling and temperature scaling (Platt et al., 1999; Guo et al., 2017), but it may be more stable when the labeled sample is small. Its guarantee is correspondingly narrower: it need not correct nonlinear miscalibration, but it is optimal within the class of affine transformations.

Let

$$\big(\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2, \qquad m^\star_{n,\text{lin}}(X) := \hat{a}_{\text{lin}}\, m(X) + \hat{b}_{\text{lin}}. \tag{13}$$

Equivalently, $m^\star_{n,\text{lin}}$ is obtained by empirical risk minimization over the affine class $\mathcal{F}_{\text{lin}}$. The associated linearly calibrated PPI estimator[4]is

$$\widehat{\psi}_{\text{lin}} := \frac{1}{n+N}\left\{\sum_{i=1}^n m^\star_{n,\text{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{lin}}(\widetilde{X}_j)\right\}. \tag{14}$$

Because $\mathcal{F}_{\text{lin}}$ contains the constant function and the identity map, the least-squares fit enforces empirical mean calibration and empirical orthogonality between the calibrated score and its residuals. Consequently, $\widehat{\psi}_{\text{lin}}$ satisfies the exact AIPW representation and the no-further-calibration property within the affine class.

More generally, one can include additional covariates in the linear regression adjustment to further improve precision, provided there is sufficient labeled data (Algorithm 3).

Proposition 3 (Linear calibration as AIPW)**.** *The normal equations for equation (13) imply*

$$\mathbb{P}_n^L\{Y - m^\star_{n,\text{lin}}(X)\} = 0 \qquad \textit{and} \qquad \mathbb{P}_n^L\big[m^\star_{n,\text{lin}}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m^\star_{n,\text{lin}}(X)$,

$$\widehat{\psi}_{\text{lin}} = \rho_n \mathbb{P}_n^L\{m^\star_{n,\text{lin}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\text{lin}}(\widetilde{X})\} + \mathbb{P}_n^L\big[\hat{w}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big]. \tag{15}$$

[4]Appendix A shows that this estimator can equivalently be implemented by balancing on $(1, m(X))$ and then forming the corresponding weighted outcome estimator.

Algorithm 2 Linearly calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{b}) \in \arg\min_{a,b} \sum_{i=1}^n \{Y_i - (aT_i + b)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a}T_i + \hat{b}$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a}\widetilde{T}_j + \hat{b}$
4: Form $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}$
6: Set $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

Algorithm 3 Linearly calibrated PPI with covariate adjustment

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{\beta}, \hat{b}) \in \arg\min_{a,\beta,b} \sum_{i=1}^n \{Y_i - (a + X_i^\top \beta + bT_i)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a} + X_i^\top \hat{\beta} + \hat{b}T_i$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a} + \widetilde{X}_j^\top \hat{\beta} + \hat{b}\widetilde{T}_j$
4: Form $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}$
6: Set $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

The next result connects linear calibration to PPI++, a closely related estimator that rescales the original score by empirical-efficiency maximization. In the mean-estimation problem, the two approaches are first-order equivalent, though they need not coincide exactly in finite samples. Intuitively, both target the same one-dimensional adjustment, namely the optimal rescaling of $m(X)$, given by the population slope λ_0 in (4) from the linear regression of Y on $m(X)$. Let $\widehat{\psi}_{++}(\widehat{\lambda}_{++})$ denote the PPI++ estimator of Angelopoulos et al. (2023b), that is, the AIPW estimator based on the score $\widehat{\lambda}_{++}m(X)$, where $\widehat{\lambda}_{++}$ is obtained by empirical efficiency maximization over $\{\lambda m(X) : \lambda \in \mathbb{R}\}$ (Rubin and van der Laan, 2008).

Proposition 4 (First-order equivalence of PPI++ and linear calibration)**.** *Assume that* $\mathrm{Var}\{m(X)\} > 0$, $\mathbb{E}_{P_0}[Y^2] < \infty$, *and* $\mathbb{E}_{P_0}[m(X)^2] < \infty$. *Then*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}}\big)\left\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\right\}.$$

Moreover, $\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}} = O_p(n^{-1/2})$, *and, under* $\rho_n \to \rho_0 \in (0,1)$,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = O_p\left(\frac{1-\rho_n}{\sqrt{\rho_n}}\,(n+N)^{-1}\right) = O_p\big((n+N)^{-1}\big).$$

In particular,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = o_p\big(M^{-1/2}\big).$$

Remark 3 (Relation to prognostic-score adjustment)**.** Linear calibration can be viewed as prognostic-score regression adjustment, a classical approach to improving precision through covariate adjustment in small-sample settings (Hansen, 2008; Rosenblum and van der Laan, 2009; Lin, 2013; Schuler et al., 2022; Balzer et al., 2024; Højbjerre-Frandsen et al., 2025). Proposition 4 shows that PPI++ has the same first-order interpretation. Linear calibration can also be viewed as a special case of TMLE (Højbjerre-Frandsen and Schuler, 2026).

Remark 4 (Platt scaling adjustment)**.** The linear-calibration idea extends beyond squared-error loss. One can instead use logistic loss, Poisson loss, or, more generally, losses from canonical-link generalized linear

models (Rosenblum and van der Laan, 2009). In particular, when $Y \in \{0,1\}$ and $m(X) \in (0,1)$ is an initial predictor, one may apply Platt scaling (Cox, 1958; Platt et al., 1999) by fitting a logistic regression of Y on the logit of $m(X)$:

$$\big(\hat{a}_{\text{platt}}, \hat{b}_{\text{platt}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^{n} \Big[-Y_i \log \sigma\big(a \operatorname{logit}(m(X_i)) + b\big) - (1-Y_i) \log\Big(1 - \sigma\big(a \operatorname{logit}(m(X_i)) + b\big)\Big)\Big], \quad (16)$$

with calibrated predictor

$$m^{\star}_{n,\text{platt}}(X) := \sigma\big(\hat{a}_{\text{platt}} \operatorname{logit}(m(X)) + \hat{b}_{\text{platt}}\big),$$

where $\sigma(t) := (1+e^{-t})^{-1}$. Platt scaling may be preferred over linear regression calibration when predicted probabilities are systematically over- or under-confident, in which case the calibration curve is often approximately sigmoidal. More generally, outcomes can be rescaled to $[0,1]$ before applying this procedure. In online settings, calibration properties of Platt scaling are developed in Gupta and Ramdas (2023).

3.4 Cross-Fitting and Cross-Calibration

In the main exposition, we assume for simplicity that the initial black-box predictor is fixed. In practice, however, it may be trained on the labeled sample itself, in which case the same labeled observations are used both to fit the predictor and to calibrate its scores. A cross-fitted version avoids this reuse by replacing the single prediction function with out-of-fold predictions and then fitting the calibrator to the resulting labeled out-of-fold scores.

1. Partition the labeled sample indices into folds $I_1, \ldots, I_K$.
2. For each fold k, train the black-box regression $m^{(-k)}$ on the labeled observations outside I_k, compute out-of-fold predictions $m^{(-k)}(X_i)$ for $i \in I_k$, and compute predictions $m^{(-k)}(\widetilde{X}_j)$ on the unlabeled sample.
3. Pool the labeled out-of-fold predictions $\{m^{(-k(i))}(X_i)\}_{i=1}^{n}$ and fit a calibrator f_n to the labeled pairs $(m^{(-k(i))}(X_i), Y_i)$.
4. Form calibrated predictions
$$m^{\star}_n(X_i) = f_n(m^{(-k(i))}(X_i))$$
and
$$m^{\star}_n(\widetilde{X}_j) = \frac{1}{K}\sum_{k=1}^{K} f_n(m^{(-k)}(\widetilde{X}_j)),$$
and then compute the plug-in estimator in equation (5).

This cross-fitted modification is the natural DML-style implementation when one wishes to reduce overfitting from using the same labeled data both to estimate the initial regression and to calibrate it for inference (Zheng and Van Der Laan, 2010; van der Laan and Rose, 2011; Chernozhukov et al., 2018a). Related procedures have been used for debiased inference in van der Laan et al. (2024b,c); Rabenseifner et al. (2025), and a related cross-calibration variant appears in van der Laan et al. (2023b).

4 Theory for isotonic-calibrated PPI

We now study the theoretical basis for *calibeating* via isotonic calibration. Our goal is to understand when post-processing an available score by isotonic regression can improve on the original score and on simpler calibration rules. To that end, we analyze the isotonic-calibrated PPI estimator $\widehat{\psi}_{\text{iso}}$, establishing asymptotic linearity, asymptotic normality, and the validity of Wald inference. We then study its efficiency, giving conditions under which it is asymptotically at least as efficient as AIPW, PPI, PPI++, and linear calibration, and conditions under which it attains the minimum asymptotic variance among regular estimators based on

$m(X)$ or on the full covariate vector X (Bickel et al., 1993; Van der Vaart, 2000). Finally, we turn to the calibration side of calibeating. We show that isotonic calibration is first-order optimal in a natural sense: once the score has been isotonically calibrated, no further post-processing based only on the fitted score can improve either its mean squared prediction error or the leading asymptotic behavior of the resulting plug-in estimator. To keep the exposition light, we focus on the non-cross-fitted estimator and treat the initial black-box score m as fixed.

4.1 Asymptotic normality, inference, and efficiency

We let f_0 denote a population isotonic calibration map for the fixed score m, defined by

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \mathbb{E}_{P_0}\big[\{Y - f(m(X))\}^2\big], \qquad m_0 = f_0 \circ m. \tag{17}$$

Thus, m_0 is the population L^2-optimal monotone transformation of the initial score $m(X)$. Because the identity map belongs to $\mathcal{F}_{\text{iso}}$, m_0 has squared-error risk no larger than that of the original score m.

Assumption 2 (Regularity conditions for isotonic calibration)**.** Assume:

(i) **Boundedness:** There exists $C_0 < \infty$ such that $|\mathbb{E}_{P_0}[Y \mid m(X)]| \le C_0$ almost surely and $\sup_x |m^\star_{n,\text{iso}}(x)| = O_p(1)$.

(ii) **Controlled tails:** The conditional error $Y - \mathbb{E}_{P_0}[Y \mid m(X)]$ is sub-Gaussian or subexponential under P_0.

Condition (i) requires only that the regression target be bounded and that the isotonic estimator remain bounded in probability. Condition (ii) allows Y to be unbounded, covering both Gaussian-like tails and somewhat heavier exponential-type tails. In particular, it holds for binary, Gaussian, Poisson, and bounded outcomes. Existing results on isotonic regression under heavy-tailed errors also suggest some robustness to violations of these conditions (Han and Wellner, 2018, 2019).

The target of the asymptotic analysis is the population isotonic limit m_0, which need not equal the true regression μ_0. Under Assumption 2, the isotonic estimator attains the classical rate

$$\|m^\star_{n,\text{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3});$$

see, for example, Chatterjee et al. (2015); Yang and Barber (2018).

Theorem 2 (Asymptotic linearity under the two-sample model)**.** *Suppose Assumptions 1 and 2 hold. Then*

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 &= \rho_n(\mathbb{P}^L_n - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ &\quad + (1-\rho_n)(\mathbb{P}^U_N - P_{0,X})\{m_0 - \psi_0\} + O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big).\end{aligned} \tag{18}$$

Thus, under the $\sqrt{M}$ scaling, $\widehat{\psi}_{\text{iso}}$ is asymptotically linear with influence pair $(D^L_{m_0}, D^U_{m_0})$ given by

$$D^L_{m_0}(X,Y) = m_0(X) - \psi_0 + \rho_0^{-1}\{Y - m_0(X)\}, \qquad D^U_{m_0}(\widetilde{X}) = m_0(\widetilde{X}) - \psi_0.$$

The next corollary records the resulting asymptotic normality and the asymptotic validity of standard Wald confidence intervals. The limiting variance of $\widehat{\psi}_{\text{iso}}$ is

$$\sigma_0^2 := \rho_0 \operatorname{Var}\{D^L_{m_0}(X,Y)\} + (1-\rho_0)\operatorname{Var}\{D^U_{m_0}(\widetilde{X})\}.$$

Corollary 1 (Asymptotic normality and Wald inference)**.** *Under the conditions of theorem 2, assume that $\sigma_0^2 > 0$. Define the empirical variance*

$$\widehat{\sigma}^2 := \rho_n \mathbb{P}^L_n[(\widehat{D}^L)^2] + (1-\rho_n)\mathbb{P}^U_N[(\widehat{D}^U)^2],$$

where

$$\widehat{D}_i^L := m^\star_{n,\mathrm{iso}}(X_i) - \widehat{\psi}_{\mathrm{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{iso}}(X_i)\}, \qquad \widehat{D}_j^U := m^\star_{n,\mathrm{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{iso}}.$$

Then

$$\frac{\sqrt{M}\,(\widehat{\psi}_{\mathrm{iso}} - \psi_0)}{\widehat{\sigma}} \xrightarrow{d} N(0,1).$$

The next corollary describes two important efficiency regimes: one in which isotonic calibration attains full semiparametric efficiency, and one in which it is optimal among estimators that use the covariates only through the one-dimensional score $m(X)$.

Corollary 2 (Efficiency regimes)**.** *Under the conditions of theorem 2, the following hold.*

(i) If $m_0 = \mu_0$ almost surely, then $\widehat{\psi}_{\mathrm{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

(ii) If $\mathbb{E}_{P_0}[Y \mid m(X)]$ is monotone increasing in $m(X)$, then $m_0(X) = \mathbb{E}_{P_0}[Y \mid m(X)]$ almost surely. Hence $\widehat{\psi}_{\mathrm{iso}}$ is efficient for ψ_0 in the reduced two-sample experiment obtained by replacing X with the scalar score $m(X)$.

Remark 5 (Practical takeaway)**.** Isotonic calibration can improve efficiency when the original score is informative but miscalibrated. The resulting estimator behaves as if $m(X)$ were replaced by its best monotone calibration $m_0(X)$. Because the isotonic class contains the identity map and all affine maps with positive slope, the isotonic-calibrated estimator has limiting variance no larger than that of any estimator that is first-order equivalent to AIPW with score $\theta\{m(X)\}$ for such a transformation θ. This includes AIPW and PPI based on $m(X)$, as well as PPI++, and linear calibration whenever the corresponding scaling or slope coefficient is nonnegative. Thus isotonic calibration is asymptotically at least as efficient, at first order, as these methods, and can be strictly more efficient when the best monotone calibration is genuinely nonlinear. More generally, if $\mathbb{E}_{P_0}[Y \mid m(X)]$ is monotone, then no regular estimator based only on $m(X)$ can improve on the isotonic-calibrated estimator at first order.

Remark 6 (When full efficiency is attained)**.** Full efficiency requires that the calibrated target recover all regression information relevant to ψ_0. This holds if $m_0 = \mu_0$ almost surely. More generally, full efficiency fails when the one-dimensional score $m(X)$ omits predictive information in X that cannot be recovered by monotone calibration.

4.2 No further first-order gain from post-processing

We now ask whether the isotonic-calibrated score can be improved by an additional post-processing step. The key observation is that $\widehat{\psi}_{\mathrm{iso}}$ depends on the covariates only through the one-dimensional fitted score $m^\star_{n,\mathrm{iso}}(X)$. It is therefore natural to study the reduced two-sample experiment

$$\mathcal{E}_{m^\star_{n,\mathrm{iso}}} := \Big(\{(m^\star_{n,\mathrm{iso}}(X_i), Y_i)\}_{i=1}^n,\ \{m^\star_{n,\mathrm{iso}}(\widetilde{X}_j)\}_{j=1}^N\Big),$$

in which the full covariate X is replaced by the scalar score $m^\star_{n,\mathrm{iso}}(X)$.

Within this reduced experiment, the natural benchmark is the estimator that replaces $m^\star_{n,\mathrm{iso}}(X)$ by the regression function $\mathbb{E}_{P_0}[Y \mid m^\star_{n,\mathrm{iso}}(X)]$. The result below shows that isotonic calibration is already first-order optimal relative to this benchmark: further post-processing of the fitted score cannot improve the leading asymptotic behavior of the resulting plug-in estimator. In practical terms, once the score has been isotonic-calibrated, additional calibration of that fitted score yields no first-order gain.

Assumption 3 (Bounded-variation score regression)**.** The map $t \mapsto \mathbb{E}_{P_0}[Y \mid m(X) = t]$ has bounded variation on the support of $m(X)$.

Assumption 3 is mild. It requires only that the population regression of Y on the initial score $m(X)$, viewed as a one-dimensional function of the score, have bounded total variation. In particular, it allows jumps and is weaker than differentiability or Lipschitz conditions.

Proposition 5 (No further first-order improvement from post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{\psi}_{\text{eff}} := \widehat{\psi}(\widehat{m}_0), \qquad \widehat{m}_0(X) := \mathbb{E}_{P_0}[Y \mid m^{\star}_{n,\text{iso}}(X)]$$

denote the benchmark estimator in the reduced experiment based on the isotonic score $m^{\star}_{n,\text{iso}}(X)$. Then

$$\|\widehat{m}_0 - m^{\star}_{n,\text{iso}}\|^2_{2,P_{0,X}} = O_p(n^{-2/3}),$$

and

$$\widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} = o_p(M^{-1/2}).$$

In particular, $\widehat{\psi}_{\text{iso}}$ and $\widehat{\psi}_{\text{eff}}$ have the same asymptotic variance.

5 Experiments

5.1 Simulation study

We begin with a controlled binary-outcome simulation designed to isolate score miscalibration. Let $S \sim N(0,1)$, define

$$\mu_0(S) = \{1 + \exp(-5S)\}^{-1},$$

and draw $Y \mid S \sim \text{Bernoulli}\{\mu_0(S)\}$. To induce substantial miscalibration, we apply an affine-biased monotone distortion on the logit scale,

$$m(X) = \text{clip}\left[-0.15 + 0.75\,\sigma\big\{0.8\,z(S) + 0.1\,z(S)^3 - 1\big\},\, 0.01,\, 0.99\right],$$

where $z(S) := \text{logit}\{\mu_0(S)\}$. This transformation preserves ranking information while introducing nonlinear distortion on the probability scale. We compare the labeled-only mean (Naive), PPI, AIPW, PPI++, an unclipped efficiency-maximized AIPW benchmark (AIPW-EM), linear calibration (LinearCal), monotone spline calibration (MonoSpline) (Ramsay, 1988; Jiang et al., 2011), isotonic calibration (IsoCal), and an adaptive selector (AutoCal) that chooses among AIPW and the main calibration rules by cross-validated empirical efficiency. Here PPI++ is the official `ppi_py` implementation, while AIPW-EM uses the same one-dimensional empirical-efficiency-maximization idea without clipping the selected scaling coefficient. The labeled sample size ranges over $n \in \{50, 100, 200, 400, 800, 1200, 2400\}$, with $N/n = 1$ and $N/n = 16$. All figures and summaries are based on 500 Monte Carlo repetitions.

figure 1 reports four diagnostics most directly tied to the theory in two representative unlabeled-sample regimes, $N/n = 1$ and $N/n = 16$: absolute bias, empirical Monte Carlo standard deviation, coverage, and relative efficiency versus PPI. All score-based AIPW-type estimators are essentially unbiased. When the original score is already well calibrated, the main practical difference is efficiency: PPI is less efficient than AIPW and the calibration-based estimators. Under meaningful miscalibration, the gains from calibration are most visible when the unlabeled sample is large. For example, at $n = 400$ and $N/n = 16$, the calibrated methods reduce RMSE from about 0.0172 for PPI to about 0.0164, and by $n = 1200$ in the same regime, RMSE is about 0.0087 for AutoCal, MonoSpline, and IsoCal, compared with 0.0092 for PPI.

The additional benchmarks sharpen the interpretation of PPI++ relative to linear calibration. In the more balanced regime, PPI++ and AIPW-EM both track LinearCal closely, consistent with the first-order equivalence developed in Section 3.3. In the large-unlabeled regime, however, AIPW-EM stays essentially on top of LinearCal while clipped PPI++ falls back toward plain PPI. Averaging over the poorly calibrated $N/n = 16$ designs, RMSE is about 0.0196 for AIPW-EM and LinearCal, versus 0.0208 for PPI++; across the 28 displayed design points, AIPW-EM improves on PPI++ in 18 and differs from LinearCal only by tiny finite-sample amounts. This points to an implementation-specific explanation: we benchmark PPI++ using the official `ppi_py` routine, which clips the optimized power-tuning coefficient λ to $[0, 1]$, whereas AIPW-EM uses the same empirical-efficiency-maximization criterion without that truncation. When the variance-optimal scaling exceeds 1, clipping pulls PPI++ back toward plain PPI. AutoCal stays close to the best fixed score-based option, while MonoSpline provides a smooth monotone alternative that is

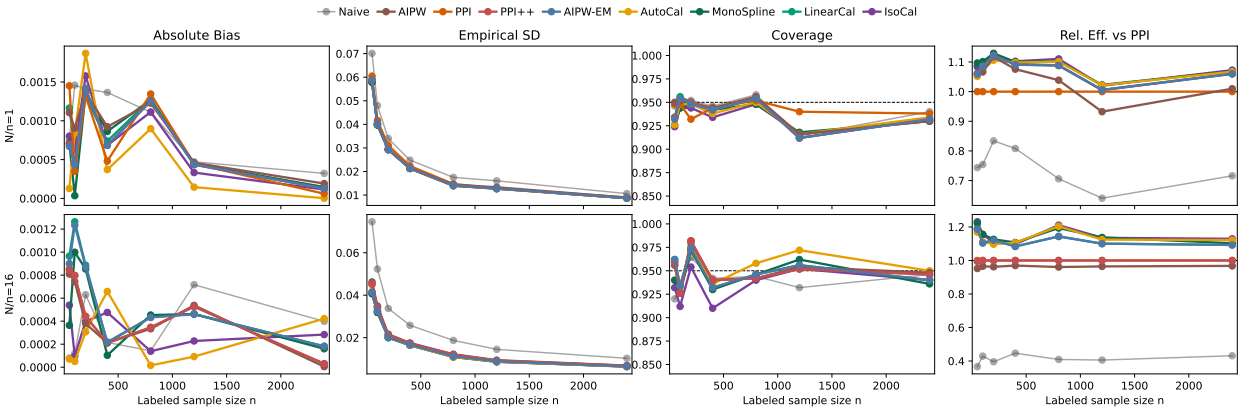

Figure 1: Synthetic simulation study with a poorly calibrated score in two unlabeled-sample regimes. The top row shows the balanced case $N/n = 1$, and the bottom row shows the large-unlabeled case $N/n = 16$. The four columns report absolute bias, empirical Monte Carlo standard deviation, coverage, and relative efficiency versus PPI, computed from empirical Monte Carlo variance. The score-based comparators now include PPI++, AIPW-EM, AutoCal, and MonoSpline in addition to PPI, AIPW, and the fixed calibration rules.

usually close to isotonic calibration in RMSE and can have slightly more stable coverage in smaller samples. Coverage is broadly reasonable overall, although the more aggressive nonlinear monotone fits can show mild undercoverage when n is small. Overall, the results support the paper's main qualitative message: little changes when the score is already well calibrated, whereas under genuine miscalibration, even simple post-hoc calibration can yield noticeable semisupervised efficiency gains.

5.2 Empirical Illustration

We evaluate the proposed calibration-based estimators on the mean-estimation benchmarks used in the original experiments of Angelopoulos et al. (2023a,b), as provided in the `ppi_py` package. Our goal is not to revisit the original benchmark comparison, but rather to understand when one-dimensional calibration of a prediction score improves semisupervised mean inference. We use the same `ppi_py` datasets and retain the labeled-sample-size grids from the original examples. For each labeled sample size n, we repeatedly split the fully labeled benchmark into a labeled subset of size n and an unlabeled subset of size N, apply all estimators to the same split, and treat the full-sample mean as the ground truth.[5] The current draft summaries are averaged over 500 random splits for each n.

Benchmarks and baselines. We study the `forest`, `galaxies`, and `census_income` semisupervised mean-estimation examples from the official benchmark suite. The retained baselines are the original labeled-only interval, the imputation benchmark on the binary-outcome datasets, PPI, the classical AIPW estimator based on the same prediction score, PPI++, and an unclipped efficiency-maximized AIPW benchmark that we denote AIPW-EM. The imputation benchmark averages model predictions over the pooled sample without a labeled residual correction. AIPW is the corresponding pooled plug-in-plus-augmentation estimator based on the original score; PPI is a less efficient variant that discards labeled covariate information from the plug-in term; PPI++ is the empirical-efficiency-maximized version of AIPW studied in the original PPI benchmark suite and implemented here through the official `ppi_py` code; and AIPW-EM applies the same one-dimensional efficiency-maximization idea through `ppi_aipw` without clipping the selected scaling coefficient.

[5] Because repeated sample splits of a fixed finite benchmark dataset are not i.i.d. draws from a superpopulation, asymptotic Wald confidence intervals can exhibit either undercoverage or overcoverage. This is not specific to our calibration extensions, but is a feature of evaluating Wald-type intervals under finite-population resampling.

We also consider four fixed calibration-based comparators that fit directly within our framework: linear calibration (`LinearCal`) (Mincer and Zarnowitz, 1969), monotone spline calibration (`MonoSpline`) (Ramsay, 1988; Jiang et al., 2011), Platt scaling (`Platt`) for the binary-outcome datasets (Platt et al., 1999), and isotonic calibration with a fixed minimum of 10 observations per bin (`IsoCal`) (Zadrozny and Elkan, 2002). We further report an adaptive selector (`AutoCal`) that chooses among AIPW, linear calibration, monotone spline calibration, and isotonic calibration by cross-validated empirical efficiency maximization (Rubin and van der Laan, 2008). For the binary-outcome datasets, we also report the Venn–Abers shrinkage variant described in Appendix C. Platt scaling and Venn–Abers are omitted for `census_income` because its outcome is not binary. To keep the main-text benchmark figure readable, however, we reserve `IsoCal` and `MonoSpline` for a calibration-focused appendix comparison in figure 3.

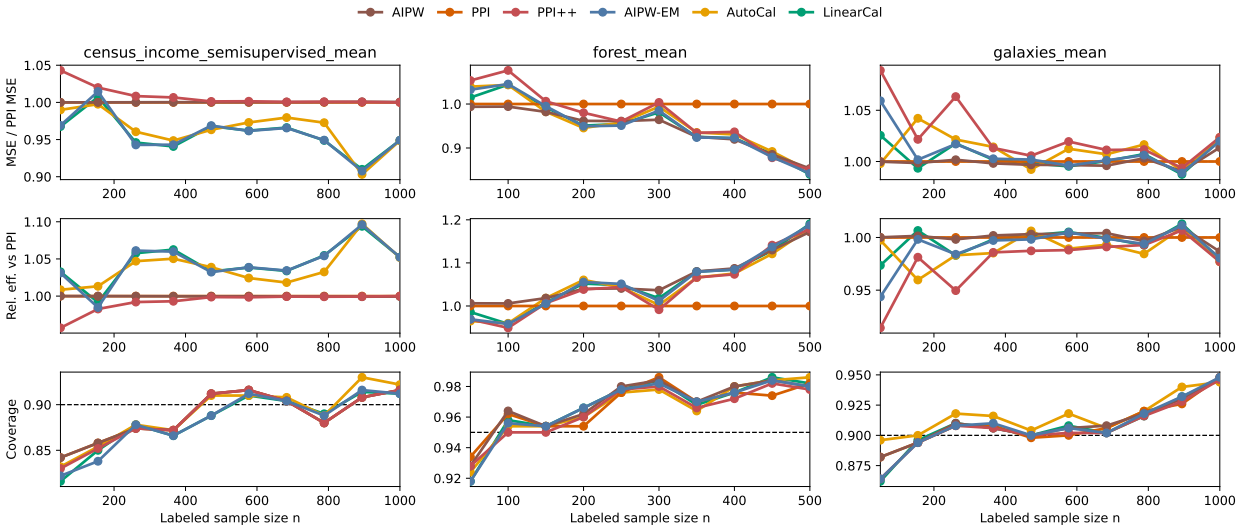

Figure 2: Main-text benchmark summary showing normalized MSE relative to PPI, relative efficiency versus PPI, and coverage across the reproduced PPI benchmarks, restricted to the primary score-based comparators: PPI, AIPW, PPI++, AIPW-EM, `AutoCal`, and `LinearCal`. In the normalized-MSE panels, values below one indicate improvement over PPI. Relative efficiency is computed from empirical Monte Carlo variance, so values above one indicate improvement over PPI. The dashed horizontal line in the coverage panels marks the nominal target coverage, which is 0.95 for `forest` and 0.90 for `galaxies` and `census_income`.

Metrics and main findings. For each estimator, we report Monte Carlo bias, empirical variance, MSE, Wald-interval coverage, and relative efficiency relative to PPI, with relative efficiency computed from empirical Monte Carlo variances. Estimated standard errors are used only to construct Wald intervals; all reported variability summaries are empirical Monte Carlo quantities.

The results suggest a simple practical picture. AIPW is the natural estimator based on the original score, and it typically matches or outperforms PPI. In `forest`, where the original score is already strong, the score-based methods perform similarly. In `galaxies`, AIPW and PPI remain nearly tied across the grid, while `AutoCal` usually stays near the top. When the unlabeled sample is much larger than the labeled sample, AIPW and PPI become nearly indistinguishable because the labeled contribution to the pooled plug-in term is negligible. This is what we see in `census_income`, where the unlabeled sample is enormous.

Post-hoc calibration is most useful when the score is meaningfully miscalibrated and the labeled sample is large enough to estimate the second-stage calibrator stably. `Census_income` provides the clearest example: `LinearCal` performs best on average, and the appendix comparison in figure 3 shows that `IsoCal` and `MonoSpline` remain broadly competitive without changing the main ranking. The added efficiency-maximization baseline sharpens the comparison with PPI++. Across the 30 dataset-size cells in figure 2, AIPW-EM has lower MSE than PPI++ in 29 and is nearly tied with `LinearCal`; averaged across all cells,

Dataset	Smaller labeled-sample regime	Larger labeled-sample regime
`forest`	AIPW and PPI are already hard to beat at the smallest n; among the efficiency-maximized one-dimensional corrections, AIPW-EM stays closer to `LinearCal` than clipped PPI++.	At larger n, all of the score-based methods cluster tightly, with only very small differences among AIPW, `LinearCal`, AIPW-EM, and PPI++.
`galaxies`	AIPW, PPI, and `AutoCal` are very close at the smallest n, while AIPW-EM and `LinearCal` are slightly more competitive than PPI++ in this noisier regime.	At larger n, the leading score-based methods remain tightly clustered, with AIPW-EM typically falling between `LinearCal` and PPI++.
`census_income`	PPI and AIPW are effectively indistinguishable because the unlabeled sample is enormous; AIPW-EM and `LinearCal` already offer modest gains at the smallest n, while PPI++ trails both.	`LinearCal` and AIPW-EM are essentially tied on average and both outperform PPI++ throughout, while PPI and AIPW remain nearly identical.

Table 1: Compact summary of the reproduced PPI benchmarks across regimes. The main empirical contrast is between small labeled-sample regimes, where simple or adaptive low-complexity corrections are safer, and larger labeled-sample regimes, where richer monotone calibration can help when the score is miscalibrated.

mean MSE is about 4.75×10^6 for AIPW-EM, 4.75×10^6 for `LinearCal`, and 5.03×10^6 for PPI++, with essentially identical coverage near 0.919. The clearest separation again appears in `census_income`, where AIPW-EM and `LinearCal` both substantially improve on PPI++; in `forest` the same ranking holds with smaller absolute gaps, and in `galaxies` the leading methods remain tightly clustered although PPI++ is still slightly weakest. This mirrors the simulations: once the empirical efficiency-maximization step is left unclipped, its finite-sample behavior moves much closer to `LinearCal`. `AutoCal` acts as a practical hedge that tends to stay close to the best fixed score-based method, while `MonoSpline` provides a smoother monotone alternative to fixed-bin isotonic calibration. Overall, these results suggest little practical reason to prefer PPI over AIPW, and they indicate that simple calibration methods such as `LinearCal` can deliver modest but repeatable gains.

6 Conclusion and closing remarks

We introduced *Calibrated Prediction-Powered Inference*, a simple framework for semisupervised mean estimation based on three steps: fit a prediction score, calibrate it on the labeled sample, and average the calibrated predictions over the pooled covariate sample. This simple plug-in procedure fits naturally within the broader literature on semiparametric debiasing and can be viewed as a semisupervised specialization of calibrated debiased machine learning (van der Laan et al., 2024c).

Our results highlight a simple message. Calibration can improve semisupervised efficiency without retraining the original model, and isotonic calibration can *calibeat* the original score and simpler post-processing rules at first order. At the same time, once the score has been isotonicly calibrated, further post-processing of that fitted score brings no additional first-order gain. Empirically, this suggests a transparent and practical workflow: when the original score is already well calibrated, calibration changes little, whereas under meaningful miscalibration, even simple linear or monotone calibration can improve efficiency while preserving a transparent plug-in form.

Positioning within existing semiparametric methods. More broadly, our results clarify that recent prediction-powered inference methods fit naturally within the standard semiparametric, debiased machine-learning, and flexible covariate-adjustment toolkit for randomized trials and missing-at-random settings. In particular, we show that PPI++ is AIPW with empirical efficiency maximization (Rubin and van der Laan, 2008) and is first-order equivalent to linear calibration and classical prognostic-score regression adjustment (Hansen, 2008; Rosenblum and van der Laan, 2009; Moore and van der Laan, 2009a; Lin, 2013). It is therefore closely related to randomized-trial approaches that learn prognostic scores from larger historical or auxiliary

data sets and then use them for regression adjustment in smaller trials (Hansen, 2008; Schuler et al., 2022; Højbjerre-Frandsen et al., 2025). This perspective suggests that semisupervised inference may benefit from closer integration with the broader missing-data, causal-inference, and debiased machine-learning literatures. These lines of work address closely related versions of the same underlying problem, so further progress may come from translational work across them, especially in adapting tools developed for randomized trials and missing-at-random settings to semisupervised problems (Smith et al., 2023).

Beyond mean estimation. These connections extend beyond the semisupervised mean-estimation setting considered here. For example, when missingness affects both outcomes and covariates and may depend on observed covariates and outcomes, as well as in extensions to right-censored time-to-event outcomes and longitudinal settings, existing semiparametric methods can often be applied directly (Robins et al., 1994; van der Laan and Robins, 2003; Bang and Robins, 2005; Moore and van der Laan, 2009b; Rose and van der Laan, 2011; Van der Laan and Rose, 2018). Likewise, there is already a well-developed literature on machine-learning-assisted estimation of local and conditional regression targets, including dose-response curves and conditional average treatment effects, that can be used to extend inference beyond marginal mean estimation (Rubin and van der Laan, 2006; Díaz and van der Laan, 2013; Kennedy et al., 2017; Bibaut and van der Laan, 2017; van der Laan et al., 2018; Westling et al., 2020; Kennedy, 2023; Foster and Syrgkanis, 2023; Luedtke and Chung, 2024; Chernozhukov et al., 2024; Butzin-Dozier et al., 2024; Zhang et al., 2025). In particular, kernel-regression-based (Bibaut and van der Laan, 2017) and isotonic-regression-based (Westling and Carone, 2020) inference can be carried out in this setting using AIPW-score-based pseudo-outcomes, as in Van Der Laan and Dudoit (2003); Rubin and van der Laan (2006); Kennedy et al. (2017), or more generally using influence functions and Neyman-orthogonal losses (Bibaut and van der Laan, 2017; Foster and Syrgkanis, 2023; Chernozhukov et al., 2024). Finally, when outcomes are not missing completely at random, naively pooling labeled and unlabeled data can bias estimation because the missingness mechanism may remain confounded by unobserved outcomes. There is substantial work on combining gold-standard and potentially biased auxiliary data more safely using semiparametric and machine-learning tools (Kallus et al., 2018; Rosenman, 2025; Dang et al., 2025; van der Laan et al., 2026). Likewise, when the unlabeled sample is selected in a data-dependent way, the problem is closely related to adaptive and selectively sampled designs, including two-phase and informative sampling (van der Laan, 2008; Chow and Chang, 2008; Rose and van der Laan, 2011; Malenica et al., 2021; Zhang and van der Laan, 2025; Liu et al., 2025). Extending these ideas to semisupervised settings is therefore a natural direction for future work.

Extensions of calibration. While we focus on calibration of regression functions and inference on means, the ideas here may extend to more general supervised learning problems defined through loss minimization, such as median and quantile regression (Noarov and Roth, 2023; Jung et al., 2021; Roth, 2022; Whitehouse et al., 2024). As a concrete example, Algorithm 3 extends our linearly calibrated mean estimator to linear regression by including the prediction score as a covariate, in the spirit of classical prognostic adjustment. Related validity results for such adjustments in generalized linear models are given by Rosenblum and van der Laan (2009). Similarly, for median and quantile regression, one could include prediction scores as covariates. More generally, when the target parameter is defined by a moment equation, as in M-estimation, one could debias or calibrate the associated augmented moment equation (van der Laan and Robins, 2003); see Appendix E. A practical challenge in this setting is that augmented loss functions may be nonconvex, which can complicate optimization. In such cases, a more practical alternative may be a multiaccuracy-style adjustment rather than explicit augmentation or calibration, as in the Efficient Plug-in Learning framework of van der Laan et al. (2024a), which addresses this issue for nonconvex Neyman-orthogonal loss functions (Foster and Syrgkanis, 2023). Debiased inference and influence-function theory for estimating equations are studied by van der Laan and Robins (2003); Chernozhukov et al. (2018a), while generic smooth functionals of M-estimands are studied by van der Laan et al. (2025a).

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A Additional background on semiparametric statistics

The next result shows that the efficient choice in Proposition 1 is not merely optimal within the AIPW class. In fact, every regular estimator[6] in the unrestricted two-sample model is asymptotically equivalent to $\widehat{\psi}(f)$ for some score function f. Thus, the AIPW family characterizes the full class of regular estimators at first order in this model.

Proposition 6 (Characterization of regular estimators)**.** *Under Assumption 1, a mean-zero pair (D^L, D^U) is the influence pair of a regular estimator of ψ_0 in the unrestricted two-sample model if and only if there exists a function $f_0 : \mathcal{X} \to \mathbb{R}$ with finite second moment, potentially depending on P_0, such that*

$$(D^L, D^U) = (D^L_{f_0}, D^U_{f_0}),$$

where D^L_f and D^U_f are given in (2)–(3). Equivalently, for every regular estimator $\widehat{\psi}^\dagger$, there exists such an f_0 for which

$$\widehat{\psi}^\dagger - \widehat{\psi}(f_0) = o_p(M^{-1/2}).$$

[6]Informally, a regular estimator is one whose large-sample behavior remains stable under small changes in the data-generating distribution (Van der Vaart, 2000).

Proof of proposition 6. If $(D^L, D^U) = (D^L_{f_0}, D^U_{f_0})$ for some f_0, then proposition 1 shows that $\widehat{\psi}(f_0)$ is asymptotically linear with influence pair (D^L, D^U). Any asymptotically linear estimator is regular (Van der Vaart, 2000), so this proves the "if" direction.

For the converse, let (D^L, D^U) be the mean-zero influence pair of a regular estimator of ψ_0. Fix bounded measurable functions $a : \mathcal{X} \to \mathbb{R}$ and $b : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ satisfying

$$\mathbb{E}_{P_{0,X}}\{a(X)\} = 0, \qquad \mathbb{E}_{P_0}\{b(X,Y) \mid X\} = 0 \qquad \text{almost surely.}$$

For t near zero, define

$$\frac{dP_{t,X}}{dP_{0,X}}(x) := \frac{\exp\{ta(x)\}}{\mathbb{E}_{P_{0,X}}[\exp\{ta(X)\}]}$$

and

$$\frac{dP_t(\cdot \mid x)}{dP_0(\cdot \mid x)}(y) := \frac{\exp\{tb(x,y)\}}{\mathbb{E}_{P_0}[\exp\{tb(X,Y)\} \mid X = x]}.$$

Let P_t be the joint law obtained by combining $P_{t,X}$ with the conditional law $P_t(\cdot \mid x)$. Then $(P_t, P_{t,X})$ remains in the unrestricted two-sample model, and its score pair at $t = 0$ is $(a + b, a)$. Because (D^L, D^U) is an influence pair for the regular estimator,

$$\frac{d}{dt}\mathbb{E}_{P_t}(Y)\bigg|_{t=0} = \rho_0 \mathbb{E}_{P_0}\left[D^L(X,Y)\{a(X) + b(X,Y)\}\right] + (1-\rho_0)\mathbb{E}_{P_{0,X}}\left[D^U(X)a(X)\right].$$

The left-hand side equals $\mathbb{E}_{P_0}[Y\{a(X) + b(X,Y)\}]$, so

$$\mathbb{E}_{P_0}[Y\{a(X) + b(X,Y)\}] = \rho_0 \mathbb{E}_{P_0}\left[D^L(X,Y)\{a(X) + b(X,Y)\}\right] + (1-\rho_0)\mathbb{E}_{P_{0,X}}\left[D^U(X)a(X)\right]. \tag{19}$$

Taking $a = 0$ in equation (19) gives

$$\mathbb{E}_{P_0}\left[\{\rho_0 D^L(X,Y) - Y\}b(X,Y)\right] = 0$$

for every bounded b with conditional mean zero given X. Now let $h : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ be bounded and set

$$b(X,Y) := h(X,Y) - \mathbb{E}_{P_0}\{h(X,Y) \mid X\}.$$

Then b satisfies the same conditional-mean-zero restriction, so

$$\mathbb{E}_{P_0}\left[\{\rho_0 D^L(X,Y) - Y\}\{h(X,Y) - \mathbb{E}_{P_0}(h \mid X)\}\right] = 0.$$

Because this holds for every bounded h, it follows that $\rho_0 D^L(X,Y) - Y$ is measurable with respect to X. Therefore there exists a square-integrable function $q : \mathcal{X} \to \mathbb{R}$ such that

$$D^L(X,Y) = \rho_0^{-1} Y + q(X).$$

Taking $b = 0$ in equation (19) and substituting the display above yields

$$\mathbb{E}_{P_{0,X}}\left[\{\rho_0 q(X) + (1-\rho_0)D^U(X)\}a(X)\right] = 0$$

for every bounded mean-zero function a. Hence the X-measurable function

$$\rho_0 q(X) + (1-\rho_0)D^U(X)$$

is almost surely constant. Since $\mathbb{E}_{P_0}[D^L(X,Y)] = 0$ and $\mathbb{E}_{P_{0,X}}[D^U(X)] = 0$,

$$\mathbb{E}_{P_{0,X}}\{q(X)\} = -\rho_0^{-1}\psi_0,$$

and therefore the constant equals

$$\rho_0 \mathbb{E}_{P_{0,X}}\{q(X)\} + (1-\rho_0)\mathbb{E}_{P_{0,X}}\{D^U(X)\} = -\psi_0.$$

Thus

$$\rho_0 q(X) + (1-\rho_0) D^U(X) = -\psi_0 \qquad \text{almost surely.}$$

Define

$$f_0(X) := \psi_0 + D^U(X).$$

Because D^U has mean zero, $\mathbb{E}_{P_{0,X}}\{f_0(X)\} = \psi_0$, so the recentered version of f_0 in equations (2) and (3) is just f_0 itself. Using $D^U(X) = f_0(X) - \psi_0$ in the previous display gives

$$q(X) = -\psi_0 - \frac{1-\rho_0}{\rho_0} f_0(X).$$

Consequently,

$$\begin{aligned} D^L(X,Y) &= \rho_0^{-1} Y + q(X) \\ &= \rho_0^{-1} Y - \psi_0 - \frac{1-\rho_0}{\rho_0} f_0(X) \\ &= f_0(X) - \psi_0 + \rho_0^{-1}\{Y - f_0(X)\}. \end{aligned}$$

Therefore

$$(D^L, D^U) = (D^L_{f_0}, D^U_{f_0}).$$

Finally, proposition 1 implies that $\widehat{\psi}(f_0)$ has the same influence pair (D^L, D^U), so subtracting the two asymptotic linear expansions gives

$$\widehat{\psi}^\dagger - \widehat{\psi}(f_0) = o_p(M^{-1/2}).$$

This proves the result. □

This characterization shows that the AIPW class is the relevant first-order benchmark in this model. Appendix A illustrates that several estimators not always written in AIPW form—including regression adjustment, balancing-weight adjustment, and TMLE—can nonetheless be viewed as finite-sample or asymptotic equivalents of AIPW estimators (Rotnitzky et al., 2025; Van Der Laan and Rubin, 2006), while sometimes offering better finite-sample performance (Porter et al., 2011; Jin, 2026; van der Laan et al., 2025a).

Remark 7 (Plug-in approaches to bias correction and TMLE)**. Regression plug-in estimators.** A particularly simple member of this first-order class is obtained through plug-in regression adjustment (Scharfstein et al., 1999; Van Der Laan and Rubin, 2006; Rosenblum and van der Laan, 2009; Rotnitzky et al., 2025). Let

$$m_n(X) = \hat{a}_{\text{lin}}\, m(X) + \hat{b}_{\text{lin}}, \qquad (\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}) = \arg\min_{a,b} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2.$$

The corresponding plug-in estimator is

$$\rho_n \mathbb{P}_n^L\{m_n(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\}.$$

For any $\lambda \in \mathbb{R}$, this estimator also admits the AIPW representation

$$\rho_n \mathbb{P}_n^L\{m_n(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\} + \lambda \mathbb{P}_n^L\big[Y - m_n(X)\big],$$

because the augmentation term $\mathbb{P}_n^L[Y - m_n(X)]$ is zero by the least-squares normal equations. Thus, when m_n converges to the best linear predictor of Y given $m(X)$, this plug-in estimator is typically at least as efficient as $\widehat{\psi}_{\text{AIPW}}$ and $\widehat{\psi}_{\text{PPI}}$, since it corresponds to an influence function with smaller or equal variance.

Balancing-weight estimators. The same estimator can also be written in terms of balancing weights. Let $w_n(X) = \hat{a}_{\text{bal}} + \hat{b}_{\text{bal}}\, m(X)$ be affine weights that balance $(1, m(X))$, in the sense that

$$\mathbb{P}_n^L\{w_n(X)\} = 1 \quad \text{and} \quad \mathbb{P}_n^L\big[m(X) w_n(X)\big] = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\}.$$

Then the corresponding balancing-weight estimator satisfies

$$\mathbb{P}_n^L\big\{w_n(X)Y\big\} = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\big[w_n(X)\{Y - m(X)\}\big].$$

In fact, the plug-in and balancing-weight estimators are algebraically equivalent (Chattopadhyay and Zubizarreta, 2023; Bruns-Smith et al., 2025):

$$\mathbb{P}_n^L\big\{w_n(X)Y\big\} = \rho_n \mathbb{P}_n^L\{m_n(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\}.$$

TMLE. More broadly, such debiased plug-in estimators can be viewed as instances of TMLE (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011; Højbjerre-Frandsen and Schuler, 2026), a general framework for constructing plug-in estimators that are debiased and, in many cases, efficient by updating nuisance components to remove first-order bias. In the present setting, either the weights or the regression model are updated so that the corresponding augmentation term vanishes. In weighting-based TMLE, the update balances on the prediction model $m(X)$ (e.g., Hejazi and van der Laan 2023); one may either include the known weight $1/\rho_n$ as an offset or balance on the intercept as well. In regression-based TMLE, the prediction model $m(X)$ may be included either as a covariate or as an offset (e.g., Lendle et al. 2015), and the intercept adjustment accounts for the known inverse missingness-probability weight $1/\rho_n$. The specific targeting step is ultimately a practical choice, provided it solves the relevant estimating equation and removes the augmentation term; different choices may also yield additional benefits, such as further bias reduction or doubly robust inference (Van der Laan, 2014; Carone et al., 2014; Díaz and van der Laan, 2017; Benkeser et al., 2017; van der Laan et al., 2021, 2024c).

B Constructing confidence intervals via the bootstrap

Confidence intervals may be constructed using the standard Wald approximation from Corollary 1, namely

$$\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\,\frac{\hat{\sigma}}{\sqrt{n+N}}.$$

As an alternative, one may use the nonparametric bootstrap and refit the isotonic calibration step within each bootstrap replicate, as proposed and theoretically studied by van der Laan et al. (2024c); see also Tang and Westling (2024). This avoids direct estimation of the influence-function variance and automatically propagates uncertainty from estimation of the calibration map.

For each bootstrap replicate $b = 1, \ldots, B$, we sample with replacement from the labeled observations $\{(X_i, Y_i)\}_{i=1}^n$ and, independently, from the unlabeled observations $\{\widetilde{X}_j\}_{j=1}^N$, obtaining bootstrap samples

$$\{(X_i^{*(b)}, Y_i^{*(b)})\}_{i=1}^n, \qquad \{\widetilde{X}_j^{*(b)}\}_{j=1}^N.$$

We then refit the isotonic calibrator on the bootstrap labeled sample:

$$\hat{f}^{*(b)} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \big\{Y_i^{*(b)} - f(m(X_i^{*(b)}))\big\}^2,$$

and define

$$m_{\text{iso}}^{*(b)}(X) := \hat{f}^{*(b)}\{m(X)\}.$$

The corresponding bootstrap replicate of the estimator is

$$\widehat{\psi}_{\text{iso}}^{*(b)} := \frac{1}{n+N}\left\{\sum_{i=1}^n m_{\text{iso}}^{*(b)}(X_i^{*(b)}) + \sum_{j=1}^N m_{\text{iso}}^{*(b)}(\widetilde{X}_j^{*(b)})\right\}.$$

After computing B bootstrap replicates, one may form a percentile interval using the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of

$$\{\widehat{\psi}_{\text{iso}}^{*(b)}\}_{b=1}^B.$$

Alternatively, one may use the bootstrap standard error

$$\widehat{\sigma}^2_{\text{boot}} := \frac{1}{B-1}\sum_{b=1}^{B}\left(\widehat{\psi}^{*(b)}_{\text{iso}} - \bar{\psi}^*_{\text{iso}}\right)^2, \qquad \bar{\psi}^*_{\text{iso}} := \frac{1}{B}\sum_{b=1}^{B}\widehat{\psi}^{*(b)}_{\text{iso}},$$

and report the normal-approximation interval

$$\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\,\widehat{\sigma}_{\text{boot}}.$$

The key implementation detail is that the isotonic regression step must be refit within each bootstrap replicate rather than held fixed. Otherwise, the resulting interval does not reflect uncertainty from estimation of the calibration map.

C Additional remarks

Remark 8 (Empirical efficiency maximization as a learning objective)**.** We review Rubin and van der Laan (2008). The empirical variance criterion from Section 2.2,

$$\sigma_n^2(f) := \rho_n \mathbb{P}_n^L\left[\left\{f(X) - \widehat{\psi}(f) + \rho_n^{-1}(Y - f(X))\right\}^2\right] + (1-\rho_n)\mathbb{P}_N^U\left[\left\{f(\widetilde{X}) - \widehat{\psi}(f)\right\}^2\right],$$

is not tied to the one-dimensional scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$. For any candidate class $\mathcal{G} \subset L^2(P_{0,X})$, one may instead define

$$\hat{f} \in \arg\min_{f\in\mathcal{G}} \sigma_n^2(f), \qquad \widehat{\psi}_{\text{eem}} := \widehat{\psi}(\hat{f}),$$

so empirical efficiency maximization learns the augmentation $f(X)$ directly.

At the population level, let

$$V(f) := \rho_0 \text{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\text{Var}\{D_f^U(\widetilde{X})\}$$

denote the asymptotic variance of $\widehat{\psi}(f)$. Using equations (2) and (3),

$$V(f) = \text{Var}(Y) + \frac{1-\rho_0}{\rho_0}\text{Var}\{Y - f(X)\}.$$

Equivalently, with $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$,

$$V(f) = \text{Var}(Y) + \frac{1-\rho_0}{\rho_0}\mathbb{E}\big[(Y - \tilde{f}(X))^2\big].$$

Thus the efficiency-optimal choice is the $L^2(P_{0,X})$-projection of μ_0 onto the normalized class $\{\tilde{f} : f \in \mathcal{G}\}$. If $\mathcal{G}$ is closed under additive constants, then minimizing $V(f)$ is equivalent to ordinary least-squares regression of Y on X over $\mathcal{G}$. In particular, rather than optimizing the empirical efficiency criterion $\sigma_n^2(f)$ directly, one can equivalently optimize the usual empirical mean-squared-error objective on the labeled sample.

This makes richer efficiency-maximizing classes immediate. For example, if $\mathcal{G} = \{b + h : h \in \mathcal{H},\ b \in \mathbb{R}\}$ for an RKHS $\mathcal{H}$ with kernel K, then a penalized version of empirical efficiency maximization is

$$(\hat{b}_\lambda, \hat{h}_\lambda) \in \arg\min_{b\in\mathbb{R},\ h\in\mathcal{H}} \left\{\mathbb{P}_n^L\big[(Y - b - h(X))^2\big] + \lambda\|h\|^2_{\mathcal{H}}\right\},$$

which is just kernel ridge regression on the labeled sample, up to the irrelevant additive constant in the variance criterion. By the representer theorem,

$$\hat{h}_\lambda(\cdot) = \sum_{i=1}^{n}\hat{\alpha}_i K(X_i, \cdot),$$

and, after centering the kernel or including the intercept explicitly,

$$\hat{\alpha} = (K_n + n\lambda I_n)^{-1}(Y - \hat{b}_\lambda \mathbf{1}_n).$$

Plugging $\hat{f}_\lambda = \hat{b}_\lambda + \hat{h}_\lambda$ into $\widehat{\psi}(\hat{f}_\lambda)$ yields the corresponding efficiency-maximized AIPW estimator.

For tuning and class selection, it is preferable to evaluate the same criterion out of fold. One may split the labeled sample into folds, fit f^{-k} on the training part of fold k, compute the held-out criterion $\sigma^2_{n,k}(f^{-k})$, and average over folds. The selected class or tuning parameter can then be used in a cross-fitted or cross-averaged AIPW estimator. This gives a direct cross-validated empirical-efficiency objective for choosing among classes such as linear, spline, isotonic, or RKHS-based regressors while preserving the usual sample-splitting protection against in-sample overfitting.

Remark 9 (Shrinkage via Venn–Abers)**.** When the labeled sample is small, isotonic calibration may be unstable. A natural alternative is Venn–Abers calibration, originally introduced for binary classification (Vovk and Petej, 2012); see van der Laan and Alaa (2024, 2025) for generalized versions beyond the binary setting. Assume $Y \in [0,1]$, or rescale otherwise. For each x, augment the labeled calibration sample with the hypothetical point $(m(x), y)$ for $y \in \{0,1\}$, fit the resulting isotonic calibrators $f_n^{(x,0)}$ and $f_n^{(x,1)}$, and form the interval-valued prediction

$$\big[f_n^{(x,0)}\{m(x)\},\ f_n^{(x,1)}\{m(x)\}\big].$$

Following Vovk and Petej (2012), one may convert this interval to a point prediction by shrinking its midpoint toward the (unadjusted) AIPW estimator:

$$m^*_{n,\mathrm{VA}}(x) := m^*_{n,\mathrm{mid}}(x) + \big\{f_n^{(x,1)}\{m(x)\} - f_n^{(x,0)}\{m(x)\}\big\}\big\{\widehat{\psi}_{AIPW} - m^*_{n,\mathrm{mid}}(x)\big\},$$

where

$$m^*_{n,\mathrm{mid}}(x) := \frac{f_n^{(x,1)}\{m(x)\} + f_n^{(x,0)}\{m(x)\}}{2}.$$

Since the shrinkage magnitude is proportional to the width of the Venn–Abers interval, this construction can be more stable than plain isotonic calibration in small samples.

Remark 10 (Relation between linear calibration, PPI, and prognostic-score adjustment)**.** Linear calibration can be viewed as prognostic-score regression adjustment, a classical approach to improving precision through flexible covariate adjustment in small-sample settings (Hansen, 2008; Rosenblum and van der Laan, 2009; Lin, 2013; Schuler et al., 2022; Balzer et al., 2024; Højbjerre-Frandsen et al., 2025). Proposition 4 shows that the PPI++ estimator admits the same first-order interpretation. Likewise, the PPI estimator $\widehat{\psi}_{\mathrm{PPI}}$ is algebraically equivalent to the unlabeled-sample plug-in estimator $\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$ based on the intercept-only adjustment

$$\hat{a}_{\mathrm{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^n \{Y_i - a - m(X_i)\}^2, \qquad m^\star_{n,\mathrm{const}}(X) := \hat{a}_{\mathrm{const}} + m(X), \tag{20}$$

which calibrates only the mean. By contrast, the standard AIPW estimator $\widehat{\psi}_{\mathrm{AIPW}}$ is equivalent to the pooled-sample mean $\rho_n \mathbb{P}_n^L\{m^\star_{n,\mathrm{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$.

D Prediction-Powered Causal Inference via Calibration (i.e., Calibrated DML)

This appendix records the direct translation of the two-sample missing-outcome setup to randomized experiments. The central point is that the semisupervised problem studied in the main text and two-arm causal inference in a randomized trial are the same statistical problem once the unobserved potential outcome in one arm is interpreted as the missing outcome. We then illustrate how our calibrated PPI estimators apply in this setting, where they arise as special cases of calibrated DML (van der Laan et al., 2024c).

We use the standard potential-outcomes notation. Each unit has two potential outcomes, $Y(1)$ under treatment and $Y(0)$ under control, and the observed outcome satisfies

$$Y = AY(1) + (1-A)Y(0),$$

where $A \in \{0,1\}$ is the treatment assignment. The mean potential outcomes are

$$\mu_1 := \mathbb{E}\{Y(1)\}, \qquad \mu_0 := \mathbb{E}\{Y(0)\},$$

and the average treatment effect is $\tau := \mu_1 - \mu_0$. In a randomized trial, identification follows from the usual causal conditions: consistency and no interference, exchangeability induced by randomization,

$$A \perp (X, Y(1), Y(0)),$$

and positivity, so that each arm has positive assignment probability.

Now partition the observed data into the two arm-specific samples

$$\mathcal{D}_{n,1} := \{(X_i, Y_i) : A_i = 1\}, \qquad \mathcal{D}_{n,0} := \{(\widetilde{X}_j, \widetilde{Y}_j) : A_j = 0\},$$

with sample sizes $n_1 := |\mathcal{D}_{n,1}|$, $n_0 := |\mathcal{D}_{n,0}|$, and $n = n_1 + n_0$. Let $\mathbb{P}_{n,a}$ denote the empirical mean over arm a, and let $\hat{\pi}_a := n_a/n$. To estimate μ_1, one treats $\mathcal{D}_{n,1}$ as the labeled sample for $Y(1)$ and uses the covariates in $\mathcal{D}_{n,0}$ as the unlabeled sample. To estimate μ_0, one reverses the roles of the two arms. Thus each mean potential outcome is a two-sample missing-outcome estimand of exactly the form analyzed in the main text.

For $a \in \{0,1\}$, let $m_a(X)$ be an arm-specific prediction score fit using only observations in arm a. The standard arm-specific AIPW estimator is

$$\hat{\mu}_a^{\mathrm{AIPW}} := \mathbb{P}_n\{m_a(X)\} + \mathbb{P}_n\left[\frac{\mathbb{1}(A=a)}{\hat{\pi}_a}\{Y - m_a(X)\}\right].$$

For $a \in \{0,1\}$, let $m^*_{n,a}(X)$ denote the calibrated version of the score $m_a(X)$. The calibrated estimator of μ_a is then

$$\hat{\mu}_a^{\mathrm{cal}} := \mathbb{P}_n\{m^*_{n,a}(X)\} = \hat{\pi}_a\, \mathbb{P}_{n,a}\{m^*_{n,a}(X)\} + (1-\hat{\pi}_a)\, \mathbb{P}_{n,1-a}\{m^*_{n,a}(X)\}. \tag{21}$$

This is exactly the pooled plug-in estimator from the two-sample formulation, with one arm contributing outcomes and the other contributing only covariates. The same estimator also has the exact AIPW form

$$\hat{\mu}_a^{\mathrm{cal}} = \mathbb{P}_n\{m^*_{n,a}(X)\} + \mathbb{P}_n\left[\frac{\mathbb{1}(A=a)}{\hat{\pi}_a}\{Y - m^*_{n,a}(X)\}\right]. \tag{22}$$

Because

$$\mathbb{P}_n\left[\frac{\mathbb{1}(A=a)}{\hat{\pi}_a}\{Y - m^*_{n,a}(X)\}\right] = \mathbb{P}_{n,a}\{Y - m^*_{n,a}(X)\},$$

the residual correction vanishes exactly whenever the calibrator includes the constant score and therefore enforces mean calibration within arm a. This is the randomized-trial analogue of the exact augmented representation in theorem 1.

The average treatment effect is then estimated by

$$\hat{\tau}^{\mathrm{cal}} := \hat{\mu}_1^{\mathrm{cal}} - \hat{\mu}_0^{\mathrm{cal}}. \tag{23}$$

Equivalently,

$$\hat{\tau}^{\mathrm{cal}} = \mathbb{P}_n\{\tilde{m}_1(X) - \tilde{m}_0(X)\} + \mathbb{P}_n\left[\frac{A}{\hat{\pi}_1}\{Y - \tilde{m}_1(X)\} - \frac{1-A}{\hat{\pi}_0}\{Y - \tilde{m}_0(X)\}\right].$$

Its influence function is the difference of the two arm-specific AIPW influences,

$$D_\tau(O) = \left\{\tilde{m}_1(X) - \mu_1 + \frac{A}{\pi_1}\big(Y - \tilde{m}_1(X)\big)\right\} - \left\{\tilde{m}_0(X) - \mu_0 + \frac{1-A}{\pi_0}\big(Y - \tilde{m}_0(X)\big)\right\}.$$

Accordingly, Wald inference for the ATE may be based on the empirical variance of the estimated influence values divided by n, or on the nonparametric bootstrap refitting the calibration step within each bootstrap sample.

The practical implementation is therefore immediate. One splits the trial into $\mathcal{D}_{n,1}$ and $\mathcal{D}_{n,0}$, treats one arm as labeled and the other as unlabeled when estimating a given mean potential outcome, fits and calibrates an arm-specific score, averages the calibrated predictions over the pooled covariate sample, and finally subtracts the two arm-specific estimates to obtain the ATE. In this way, the semisupervised mean estimators in the main text become direct estimators of mean potential outcomes and treatment effects in randomized experiments.

E General Moment Equations Under Missing at Random

We briefly review the general AIPW class for moment-equation targets under missing at random, following Robins et al. (1994, 1995); Robins and Rotnitzky (1995); van der Laan and Robins (2003). Let

$$O = (X, R, RY),$$

where $R \in \{0, 1\}$ indicates whether Y is observed, and suppose

$$R \perp Y \mid X, \qquad \pi_0(X) := \mathbb{P}(R = 1 \mid X) > 0 \text{ a.s.}$$

Let the target $\theta_0 \in \Theta \subseteq \mathbb{R}^d$ be identified by

$$\mathbb{E}_{P_0}[\varphi(Y, \theta_0)] = 0,$$

where $\varphi : \mathcal{Y} \times \Theta \to \mathbb{R}^d$ is a known identifying function. Define

$$A_0 := \partial_\theta \mathbb{E}_{P_0}[\varphi(Y, \theta)]\big|_{\theta=\theta_0},$$

and assume A_0 is nonsingular.

Theorem 3 (AIPW class for general moment equations under MAR)**.** *For any square-integrable function $m : \mathcal{X} \times \Theta \to \mathbb{R}^d$, define*

$$\psi(O; \theta, m, \pi_0) := m(X, \theta) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}. \tag{24}$$

Then

$$\mathbb{E}[\psi(O; \theta_0, m, \pi_0)] = 0.$$

Hence the equation

$$\mathbb{P}_n \psi(O; \hat{\theta}, m, \pi_0) = 0$$

defines an unbiased AIPW estimating equation for θ_0.

Moreover, let $\hat{\theta}$ be a regular $\sqrt{n}$-consistent solution of this equation, and assume that $\theta \mapsto \mathbb{E}[\psi(O; \theta, m, \pi_0)]$ is differentiable at θ_0 with derivative A_0, and that

$$(\mathbb{P}_n - P_0)\big[\psi(O; \hat{\theta}, m, \pi_0) - \psi(O; \theta_0, m, \pi_0)\big] = o_p\big(n^{-1/2} + \|\hat{\theta} - \theta_0\|\big).$$

Then $\hat{\theta}$ has influence function

$$D_m(O) = -A_0^{-1}\psi(O; \theta_0, m, \pi_0) = -A_0^{-1}\left[m(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m(X, \theta_0)\}\right]. \tag{25}$$

Thus the class of regular Z-estimators arising from (24) is indexed by the choice of augmentation function m. The efficient choice in the full MAR model is

$$m_0(X, \theta_0) := \mathbb{E}[\varphi(Y, \theta_0) \mid X], \tag{26}$$

which yields the efficient influence function

$$D_{\text{eff}}(O) = -A_0^{-1}\left[m_0(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m_0(X, \theta_0)\}\right]. \tag{27}$$

Proof of theorem 3. For any $\theta \in \Theta$,

$$\begin{aligned}
\mathbb{E}\{\psi(O; \theta, m, \pi_0)\} &= \mathbb{E}\left[m(X, \theta) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}\left[m(X, \theta) + \frac{\mathbb{E}(R \mid X, Y)}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}\left[m(X, \theta) + \frac{\mathbb{E}(R \mid X)}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}\big[\varphi(Y, \theta)\big],
\end{aligned}$$

where the third line uses $R \perp Y \mid X$. Evaluating at $\theta = \theta_0$ gives

$$\mathbb{E}\{\psi(O;\theta_0,m,\pi_0)\} = \mathbb{E}\big[\varphi(Y,\theta_0)\big] = 0.$$

Let

$$\Psi_m(\theta) := \mathbb{E}\{\psi(O;\theta,m,\pi_0)\}.$$

The previous calculation shows that $\Psi_m(\theta) = \mathbb{E}[\varphi(Y,\theta)]$ for every θ, so

$$\partial_\theta \Psi_m(\theta)\big|_{\theta=\theta_0} = A_0.$$

Because $\hat{\theta}$ solves the empirical estimating equation,

$$0 = \mathbb{P}_n\psi(O;\hat{\theta},m,\pi_0) = (\mathbb{P}_n - P_0)\psi(O;\theta_0,m,\pi_0) + \{\Psi_m(\hat{\theta}) - \Psi_m(\theta_0)\} + (\mathbb{P}_n - P_0)\big[\psi(O;\hat{\theta},m,\pi_0) - \psi(O;\theta_0,m,\pi_0)\big].$$

By differentiability of Ψ_m at θ_0,

$$\Psi_m(\hat{\theta}) - \Psi_m(\theta_0) = A_0(\hat{\theta} - \theta_0) + o(\|\hat{\theta} - \theta_0\|).$$

Since $\hat{\theta}$ is $\sqrt{n}$-consistent, the previous display and the stated remainder condition imply

$$A_0(\hat{\theta} - \theta_0) = -(\mathbb{P}_n - P_0)\psi(O;\theta_0,m,\pi_0) + o_p(n^{-1/2}).$$

Multiplying by $-A_0^{-1}$ gives

$$\hat{\theta} - \theta_0 = (\mathbb{P}_n - P_0)D_m(O) + o_p(n^{-1/2}),$$

with D_m given in equation (25). This proves the influence-function formula.

Now define

$$m_0(X,\theta_0) := \mathbb{E}\{\varphi(Y,\theta_0) \mid X\}, \qquad \delta_m(X) := m(X,\theta_0) - m_0(X,\theta_0).$$

Then

$$D_m(O) = D_{\text{eff}}(O) - A_0^{-1}\left(1 - \frac{R}{\pi_0(X)}\right)\delta_m(X).$$

Fix any $u \in \mathbb{R}^d$. Because $\delta_m(X)$ is measurable with respect to X,

$$\mathbb{E}\left[\left(1 - \frac{R}{\pi_0(X)}\right)\delta_m(X)\,\middle|\, X\right] = 0.$$

Also, using $R \perp Y \mid X$ and $\mathbb{E}[\varphi(Y,\theta_0) - m_0(X,\theta_0) \mid X] = 0$,

$$\mathbb{E}\left[\frac{R}{\pi_0(X)}\left(1 - \frac{R}{\pi_0(X)}\right)\{\varphi(Y,\theta_0) - m_0(X,\theta_0)\}\,\middle|\, X\right] = 0.$$

Therefore the cross-term vanishes:

$$\mathbb{E}\left[\{u^\top D_{\text{eff}}(O)\}\left(1 - \frac{R}{\pi_0(X)}\right)\{u^\top A_0^{-1}\delta_m(X)\}\right] = 0.$$

Moreover,

$$\mathbb{E}\left[\left(1 - \frac{R}{\pi_0(X)}\right)^2\,\middle|\, X\right] = \frac{1 - \pi_0(X)}{\pi_0(X)}.$$

Expanding the variance of $u^\top D_m(O)$ therefore yields

$$\text{Var}\{u^\top D_m(O)\} = \text{Var}\{u^\top D_{\text{eff}}(O)\} + \mathbb{E}\left[\frac{1 - \pi_0(X)}{\pi_0(X)}\{u^\top A_0^{-1}\delta_m(X)\}^2\right].$$

The second term is nonnegative and equals zero if and only if $\delta_m(X) = 0$ almost surely. Hence $m_0(X,\theta_0)$ minimizes the asymptotic variance of every linear contrast, so D_{eff} is the efficient influence function within this class. The missing-data efficiency results of Robins et al. (1994, 1995); Robins and Rotnitzky (1995); van der Laan and Robins (2003) identify this same influence function as the semiparametric efficient influence function in the full MAR model. □

Corollary 3 (Two-sample/PPI form)**.** *Suppose the labeled and unlabeled samples arise from the semisupervised model with constant labeling probability ρ_0, and let $S = f(X)$ be a chosen score. Replacing X by S gives the reduced AIPW class*

$$\psi(O;\theta,m,\rho_0) = m(S,\theta) + \frac{R}{\rho_0}\{\varphi(Y,\theta) - m(S,\theta)\}.$$

In the two-sample notation used in the main text, the empirical estimating equation based on the known labeled fraction $\rho_n = n/(n+N)$ can be written as

$$0 = \rho_n \mathbb{P}_n^L\{m(S,\theta)\} + (1-\rho_n)\mathbb{P}_N^U\{m(S,\theta)\} + \mathbb{P}_n^L\{\varphi(Y,\theta) - m(S,\theta)\}.$$

The efficient choice within the reduced score model is

$$m_0(S,\theta_0) = \mathbb{E}[\varphi(Y,\theta_0) \mid S].$$

Proof. Because the semisupervised design has constant labeling probability, R is independent of (X,Y), hence also independent of Y given $S = f(X)$, and

$$\mathbb{P}(R = 1 \mid S) = \rho_0.$$

Applying theorem 3 with covariate S and observation probability $\pi_0(S) \equiv \rho_0$ gives the displayed reduced AIPW class and the efficient choice

$$m_0(S,\theta_0) = \mathbb{E}[\varphi(Y,\theta_0) \mid S].$$

At the sample level, the two-sample implementation replaces the population labeling probability ρ_0 by the known design fraction $\rho_n = n/(n+N)$. To rewrite the empirical equation in the two-sample notation, note that

$$\mathbb{P}_M\{m(S,\theta)\} = \rho_n \mathbb{P}_n^L\{m(S,\theta)\} + (1-\rho_n)\mathbb{P}_N^U\{m(S,\theta)\},$$

where $\mathbb{P}_M$ denotes the empirical mean over the pooled sample. Also,

$$\mathbb{P}_M\left[\frac{R}{\rho_n}\{\varphi(Y,\theta) - m(S,\theta)\}\right] = \mathbb{P}_n^L\{\varphi(Y,\theta) - m(S,\theta)\},$$

because $R = 1$ on labeled observations and $R = 0$ on unlabeled observations. Adding the two displays yields exactly

$$0 = \rho_n \mathbb{P}_n^L\{m(S,\theta)\} + (1-\rho_n)\mathbb{P}_N^U\{m(S,\theta)\} + \mathbb{P}_n^L\{\varphi(Y,\theta) - m(S,\theta)\}.$$

□

F Proof of the representation theorem

Proof of theorem 1. Because equation (8) holds for every $h \in \mathcal{F}$, any empirical balancing weight $\widehat{w}(m_n^\star(X)) \in \mathcal{F}$ satisfies

$$\mathbb{P}_n^L\big[\widehat{w}(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0.$$

Adding this zero term to the plug-in estimator equation (5) yields equation (9).

For the second claim, since the identity map belongs to $\mathcal{F}$, taking $f(t) = t$ in equation (10) gives

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\, m_n^\star(X)\big] = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} = \widehat{\psi}_{\mathrm{cal}}.$$

Since $\widehat{w}_n^* \in \mathcal{F}$, equation (8) also gives

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0.$$

Therefore,

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))Y\big] = \mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\, m_n^\star(X)\big] + \mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = \widehat{\psi}_{\mathrm{cal}},$$

which is equation (11). □

Lemma 1 (Blockwise residual orthogonality)**.** *Let $B_1, \dots, B_J$ be the pooled adjacent violator blocks of the isotonic fit, and let $m^\star_{n,\mathrm{iso}}(X_i) = c_j$ for $i \in B_j$. Then*

$$\sum_{i \in B_j} (Y_i - c_j) = 0, \qquad j = 1, \dots, J.$$

Consequently, for every measurable function $h : \mathbb{R} \to \mathbb{R}$ such that $h(m^\star_{n,\mathrm{iso}}(X_i))$ is finite for all i,

$$\sum_{i=1}^{n} h(m^\star_{n,\mathrm{iso}}(X_i))\{Y_i - m^\star_{n,\mathrm{iso}}(X_i)\} = 0.$$

Proof. The fitted values from isotonic regression are constant on pooled adjacent violator blocks. The KKT conditions imply that residuals sum to zero on each block. Since $h(m^\star_{n,\mathrm{iso}}(X_i)) = h(c_j)$ for all $i \in B_j$,

$$\sum_{i=1}^{n} h(m^\star_{n,\mathrm{iso}}(X_i))\{Y_i - m^\star_{n,\mathrm{iso}}(X_i)\} = \sum_{j=1}^{J} h(c_j) \sum_{i \in B_j} (Y_i - c_j) = 0.$$

□

Proof of proposition 2. The orthogonality claim equation (12) is exactly the conclusion of Lemma 1. For the risk comparison, note that the identity map belongs to $\mathcal{F}_{\mathrm{iso}}$. Since $\hat{f}$ minimizes the empirical squared loss over $\mathcal{F}_{\mathrm{iso}}$,

$$\mathbb{P}_n^L\big[(Y - m^\star_{n,\mathrm{iso}}(X))^2\big] \le \mathbb{P}_n^L\big[(Y - \hat{f}_{\mathrm{id}}(m(X)))^2\big] = \mathbb{P}_n^L\big[(Y - m(X))^2\big],$$

where $\hat{f}_{\mathrm{id}}(t) = t$. □

Proof of proposition 3. The normal equations for equation (13) are

$$\mathbb{P}_n^L\{Y - m^\star_{n,\mathrm{lin}}(X)\} = 0 \qquad \text{and} \qquad \mathbb{P}_n^L\big[m(X)\{Y - m^\star_{n,\mathrm{lin}}(X)\}\big] = 0.$$

Because $m^\star_{n,\mathrm{lin}}(X) = \hat{a}_{\mathrm{lin}} m(X) + \hat{b}_{\mathrm{lin}}$, these imply

$$\mathbb{P}_n^L\big[m^\star_{n,\mathrm{lin}}(X)\{Y - m^\star_{n,\mathrm{lin}}(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m^\star_{n,\mathrm{lin}}(X)$,

$$\mathbb{P}_n^L\big[\hat{w}(X)\{Y - m^\star_{n,\mathrm{lin}}(X)\}\big] = a\, \mathbb{P}_n^L\{Y - m^\star_{n,\mathrm{lin}}(X)\} + b\, \mathbb{P}_n^L\big[m^\star_{n,\mathrm{lin}}(X)\{Y - m^\star_{n,\mathrm{lin}}(X)\}\big] = 0.$$

Adding this zero term to equation (14) yields equation (15). □

G Proofs of first-order equivalence between PPI++ and linear calibration

Recall that

$$(\hat{a}_{\mathrm{lin}}, \hat{b}_{\mathrm{lin}}) \in \arg\min_{a,b \in \mathbb{R}} \sum_{i=1}^{n} \{Y_i - a\, m(X_i) - b\}^2,$$

and define the corresponding linear-calibration estimator by

$$\widehat{\psi}_{\mathrm{lin}} = \rho_n \mathbb{P}_n^L\{\hat{a}_{\mathrm{lin}} m(X) + \hat{b}_{\mathrm{lin}}\} + (1 - \rho_n)\mathbb{P}_N^U\{\hat{a}_{\mathrm{lin}} m(\widetilde{X}) + \hat{b}_{\mathrm{lin}}\}.$$

Also let $\widehat{\psi}_{++}$ denote the PPI++ estimator obtained by minimizing the empirical influence-function variance over the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, and let $\hat{\lambda}_{++}$ be the resulting selected coefficient. Define

$$\Delta_n := \mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X),$$

and

$$a_0 := \frac{\mathrm{Cov}\{Y, m(X)\}}{\mathrm{Var}\{m(X)\}}.$$

Proof of Proposition 4. The normal equations for the linear regression fit are

$$\mathbb{P}_n^L\{Y - \hat{a}_{\text{lin}} m(X) - \hat{b}_{\text{lin}}\} = 0$$

and

$$\mathbb{P}_n^L\big[m(X)\{Y - \hat{a}_{\text{lin}} m(X) - \hat{b}_{\text{lin}}\}\big] = 0.$$

The first equation gives

$$\hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X).$$

Substituting this into the second equation yields

$$\hat{a}_{\text{lin}} = \frac{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))(Y - \mathbb{P}_n^L Y)\big]}{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big]}.$$

Hence $\hat{a}_{\text{lin}}$ is the ordinary least-squares slope, and under finite second moments,

$$\hat{a}_{\text{lin}} = a_0 + O_p(n^{-1/2}).$$

Using

$$\hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X),$$

we obtain

$$\begin{aligned}
\widehat{\psi}_{\text{lin}} &= \rho_n \mathbb{P}_n^L\{\hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{a}_{\text{lin}} m(\widetilde{X}) + \hat{b}_{\text{lin}}\} \\
&= \hat{b}_{\text{lin}} + \hat{a}_{\text{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X) + \hat{a}_{\text{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{a}_{\text{lin}}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{a}_{\text{lin}}\Delta_n.
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\widehat{\psi}_{++}(\hat{\lambda}_{++}) &= \rho_n \mathbb{P}_n^L\{\hat{\lambda}_{++} m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{\lambda}_{++} m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - \hat{\lambda}_{++} m(X)\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{\lambda}_{++}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{\lambda}_{++}\Delta_n.
\end{aligned}$$

Therefore,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\text{lin}}\big)\Delta_n.$$

Under the empirical PPI++ variance criterion over the class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, the selected coefficient satisfies

$$\hat{\lambda}_{++} = \frac{\mathbb{P}_n^L\big[(Y - \mathbb{P}_n^L Y)(m(X) - \mathbb{P}_n^L m(X))\big]}{(1-\rho_n)\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big] + \rho_n \mathbb{P}_N^U\big[(m(\widetilde{X}) - \mathbb{P}_N^U m(\widetilde{X}))^2\big]}.$$

Its population limit is

$$\lambda_0 = \frac{\text{Cov}\{Y, m(X)\}}{\text{Var}\{m(X)\}} = a_0,$$

so standard M-estimation arguments yield

$$\hat{\lambda}_{++} = a_0 + O_p(n^{-1/2}).$$

Combining this with $\hat{a}_{\text{lin}} = a_0 + O_p(n^{-1/2})$ gives

$$\hat{\lambda}_{++} - \hat{a}_{\text{lin}} = O_p(n^{-1/2}).$$

Next, since $\mathbb{P}_n^L m(X)$ and $\mathbb{P}_N^U m(\widetilde{X})$ are sample means under the same marginal law of X,

$$\Delta_n = O_p(n^{-1/2} + N^{-1/2}).$$

Under the standing assumption $\rho_n = n/(n+N) \to \rho_0 \in (0,1)$, we may write

$$n^{-1/2} = \rho_n^{-1/2}(n+N)^{-1/2}, \qquad N^{-1/2} = (1-\rho_n)^{-1/2}(n+N)^{-1/2},$$

and hence

$$\Delta_n = O_p\Big(\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1/2}\Big).$$

Therefore,

$$\begin{aligned}
\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} &= (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}}\big)\Delta_n \\
&= (1-\rho_n)\, O_p(n^{-1/2})\, O_p\Big(\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1/2}\Big) \\
&= O_p\Big((1-\rho_n)\rho_n^{-1/2}\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1}\Big).
\end{aligned}$$

Since

$$(1-\rho_n)\rho_n^{-1/2}\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\} = \frac{1-\rho_n}{\rho_n} + \sqrt{\frac{1-\rho_n}{\rho_n}},$$

It follows that

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = O_p\Bigg(\Bigg\{\frac{1-\rho_n}{\rho_n} + \sqrt{\frac{1-\rho_n}{\rho_n}}\Bigg\}(n+N)^{-1}\Bigg).$$

In particular, this implies the simpler bound

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = O_p\Bigg(\frac{1-\rho_n}{\rho_n}(n+N)^{-1}\Bigg).$$

Therefore, under $\rho_n \to \rho_0 \in (0,1)$,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = O_p\big((n+N)^{-1}\big) = o_p\big((n+N)^{-1/2}\big).$$

This proves the claim. □

Proposition 7 (PPI and AIPW as intercept-only calibration)**.** *Let*

$$\hat{a}_{\mathrm{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a - m(X_i)\}^2, \qquad m^\star_{n,\mathrm{const}}(X) := \hat{a}_{\mathrm{const}} + m(X).$$

Then

$$\hat{a}_{\mathrm{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Consequently, the PPI estimator

$$\widehat{\psi}_{\mathrm{PPI}} := \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\mathrm{PPI}} = \mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(\widetilde{X})\},$$

while the standard AIPW estimator

$$\widehat{\psi}_{\mathrm{AIPW}} := \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\mathrm{AIPW}} = \rho_n \mathbb{P}_n^L\{m^\star_{n,\mathrm{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(\widetilde{X})\}.$$

Hence PPI is the unlabeled-only plug-in estimator based on the mean-calibrated score $m^\star_{n,\mathrm{const}}$, whereas AIPW is the corresponding pooled plug-in estimator.

Proof. The objective in the definition of $\hat{a}_{\text{const}}$ is quadratic in a, and its first-order condition is

$$0 = \mathbb{P}_n^L\{Y - a - m(X)\}.$$

Therefore

$$\hat{a}_{\text{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Using $m^\star_{n,\text{const}}(\widetilde{X}) = m(\widetilde{X}) + \hat{a}_{\text{const}}$, we obtain

$$\mathbb{P}_N^U\{m^\star_{n,\text{const}}(\widetilde{X})\} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \hat{a}_{\text{const}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\} = \widehat{\psi}_{\text{PPI}}.$$

Likewise,

$$\begin{aligned}
\rho_n\mathbb{P}_n^L\{m^\star_{n,\text{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\text{const}}(\widetilde{X})\} &= \rho_n\mathbb{P}_n^L\{m(X) + \hat{a}_{\text{const}}\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X}) + \hat{a}_{\text{const}}\} \\
&= \rho_n\mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \hat{a}_{\text{const}} \\
&= \rho_n\mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\} \\
&= \widehat{\psi}_{\text{AIPW}}.
\end{aligned}$$

This proves the claim. □

H Proof of Proposition 1

Proof of proposition 1. Let

$$c := \psi_0 - \mathbb{E}_{P_{0,X}}\{f(X)\}.$$

Because c is constant,

$$\widehat{\psi}(f + c) = \widehat{\psi}(f) + \rho_n c + (1-\rho_n)c - c = \widehat{\psi}(f).$$

Therefore $\widehat{\psi}(f) = \widehat{\psi}(\tilde{f})$, where

$$\tilde{f}(X) = f(X) + c = f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0.$$

By construction,

$$\mathbb{E}_{P_{0,X}}\{\tilde{f}(X)\} = \psi_0.$$

Using $\mathbb{E}_{P_0}[Y] = \psi_0$ and $\mathbb{E}_{P_0}\{\tilde{f}(X)\} = \mathbb{E}_{P_{0,X}}\{\tilde{f}(X)\} = \psi_0$, we obtain

$$\begin{aligned}
\widehat{\psi}(f) - \psi_0 &= \widehat{\psi}(\tilde{f}) - \psi_0 \\
&= \rho_n(\mathbb{P}_n^L - P_0)\{\tilde{f}(X)\} + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{\tilde{f}(\widetilde{X})\} + (\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\} \\
&= \rho_n(\mathbb{P}_n^L - P_0)\Big\{\tilde{f}(X) - \psi_0 + \rho_n^{-1}\big(Y - \tilde{f}(X)\big)\Big\} \\
&\quad + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\big\{\tilde{f}(\widetilde{X}) - \psi_0\big\}.
\end{aligned}$$

Define

$$D^L_{f,n}(X,Y) := \tilde{f}(X) - \psi_0 + \rho_n^{-1}\{Y - \tilde{f}(X)\}, \qquad D^U_f(\widetilde{X}) := \tilde{f}(\widetilde{X}) - \psi_0.$$

Then

$$\widehat{\psi}(f) - \psi_0 = \frac{1}{M}\sum_{i=1}^n D^L_{f,n}(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^N D^U_f(\widetilde{X}_j).$$

Next,

$$D^L_{f,n}(X,Y) - D^L_f(X,Y) = (\rho_n^{-1} - \rho_0^{-1})\{Y - \tilde{f}(X)\}.$$

Since $\mathbb{E}_{P_0}\{Y - \tilde{f}(X)\} = 0$, it follows that

$$\frac{1}{M}\sum_{i=1}^{n}\{D_{f,n}^L(X_i, Y_i) - D_f^L(X_i, Y_i)\} = \rho_n(\mathbb{P}_n^L - P_0)\Big[(\rho_n^{-1} - \rho_0^{-1})\{Y - \tilde{f}(X)\}\Big]$$
$$= \left(1 - \frac{\rho_n}{\rho_0}\right)(\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\}.$$

Because Y and $f(X)$ have finite second moments, $\tilde{f}(X)$ also has finite second moment, so

$$(\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\} = O_p(n^{-1/2}) = O_p(M^{-1/2}).$$

Since $\rho_n \to \rho_0 \in (0, 1)$ by Assumption 1,

$$\frac{1}{M}\sum_{i=1}^{n}\{D_{f,n}^L(X_i, Y_i) - D_f^L(X_i, Y_i)\} = o_p(M^{-1/2}).$$

Therefore

$$\widehat{\psi}(f) - \psi_0 = \frac{1}{M}\sum_{i=1}^{n} D_f^L(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^{N} D_f^U(\widetilde{X}_j) + o_p(M^{-1/2}),$$

which proves the stated asymptotic linearity. The mean-zero property of D_f^L and D_f^U follows from

$$\mathbb{E}_{P_0}[Y] = \mathbb{E}_{P_0}\{\tilde{f}(X)\} = \psi_0.$$

For the asymptotic variance, write

$$\delta(X) := \tilde{f}(X) - \mu_0(X), \qquad \varepsilon := Y - \mu_0(X).$$

Then

$$\mathbb{E}_{P_0}\{\delta(X)\} = 0, \qquad \mathbb{E}_{P_0}(\varepsilon \mid X) = 0, \qquad \mathbb{E}_{P_0}(\varepsilon^2) = \mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big].$$

Also,

$$D_f^U(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0 + \delta(\widetilde{X}),$$

and, since $Y - \tilde{f}(X) = \varepsilon - \delta(X)$,

$$\begin{aligned} D_f^L(X, Y) &= \tilde{f}(X) - \psi_0 + \rho_0^{-1}\{Y - \tilde{f}(X)\} \\ &= \mu_0(X) - \psi_0 + \delta(X) + \rho_0^{-1}\{\varepsilon - \delta(X)\} \\ &= \mu_0(X) - \psi_0 + \rho_0^{-1}\varepsilon + \big(1 - \rho_0^{-1}\big)\delta(X). \end{aligned}$$

Set

$$A(X) := \mu_0(X) - \psi_0, \qquad B(X) := \delta(X), \qquad C(X, Y) := \rho_0^{-1}\varepsilon.$$

Then $\mathbb{E}_{P_0}\{C(X, Y) \mid X\} = 0$, so $C(X, Y)$ is uncorrelated with both $A(X)$ and $B(X)$. Hence

$$\mathrm{Var}\{D_f^L(X, Y)\} = \mathrm{Var}\big\{A(X) + \big(1 - \rho_0^{-1}\big)B(X)\big\} + \mathrm{Var}\{C(X, Y)\},$$

while

$$\mathrm{Var}\{D_f^U(\widetilde{X})\} = \mathrm{Var}\{A(\widetilde{X}) + B(\widetilde{X})\}.$$

Therefore

$$\begin{aligned} &\rho_0\mathrm{Var}\{D_f^L(X, Y)\} + (1 - \rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\} \\ &= \rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1 - \rho_0)\mathrm{Var}(A + B) + \rho_0\mathrm{Var}(C). \end{aligned}$$

Expanding the two variances involving A and B gives

$$\begin{aligned}
&\rho_0 \mathrm{Var}\big\{A + \big(1-\rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) \\
&= \{\rho_0 + (1-\rho_0)\}\mathrm{Var}(A) \\
&\quad + \Big\{\rho_0\big(1-\rho_0^{-1}\big)^2 + (1-\rho_0)\Big\}\,\mathrm{Var}(B) \\
&\quad + 2\,\big\{\rho_0\big(1-\rho_0^{-1}\big) + (1-\rho_0)\big\}\,\mathrm{Cov}(A,B).
\end{aligned}$$

Now

$$1-\rho_0^{-1} = -\frac{1-\rho_0}{\rho_0},$$

so

$$\rho_0\big(1-\rho_0^{-1}\big)^2 + (1-\rho_0) = \frac{(1-\rho_0)^2}{\rho_0} + (1-\rho_0) = \frac{1-\rho_0}{\rho_0},$$

and

$$\rho_0\big(1-\rho_0^{-1}\big) + (1-\rho_0) = -(1-\rho_0) + (1-\rho_0) = 0.$$

Thus

$$\rho_0 \mathrm{Var}\big\{A + \big(1-\rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) = \mathrm{Var}(A) + \frac{1-\rho_0}{\rho_0}\mathrm{Var}(B).$$

Also,

$$\rho_0 \mathrm{Var}(C) = \rho_0 \cdot \rho_0^{-2}\mathbb{E}_{P_0}(\varepsilon^2) = \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big].$$

Since $\mathbb{E}_{P_0}\{B(X)\} = 0$,

$$\mathrm{Var}(B) = \mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

Combining the preceding displays yields

$$\rho_0 \mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\} = \mathrm{Var}\{\mu_0(X)\} + \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big] + \frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

This expression is minimized when $\tilde{f}(X) = \mu_0(X)$ almost surely, in particular at $f = \mu_0$. Standard semiparametric characterization results for missing-data models imply that this minimum is the semiparametric efficiency bound in the unrestricted two-sample model; see, for example, Bickel et al. (1993); Robins et al. (1995); Robins and Rotnitzky (1995); van der Laan and Robins (2003); Tsiatis (2006). This completes the proof. □

Lemma 2 (Bounded variation after isotonic post-processing)**.** *Suppose Assumption 3 holds. Let $T := m(X)$ and $S_n := m^\star_{n,\mathrm{iso}}(X)$. Define*

$$\eta(t) := \mathbb{E}_{P_0}[Y \mid T = t], \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s].$$

If $t \mapsto \eta(t)$ has total variation at most V, then, conditional on the labeled sample, the map $s \mapsto \eta_n(s)$ has total variation at most V. Moreover, $|\eta_n(s)| \le C_0$ on the support of S_n.

Proof. We adapt the proof of Lemma 6 in van der Laan et al. (2023b). Let $S_n = f_n(T)$, where f_n is the fitted isotonic map. Conditional on the labeled sample, f_n is nondecreasing and piecewise constant. Hence, for each s in the support of S_n, there exists an interval $B_s \subset \mathbb{R}$ such that

$$\{S_n = s\} = \{T \in B_s\}.$$

Therefore

$$\eta_n(s) = \mathbb{E}_{P_0}[Y \mid S_n = s] = \mathbb{E}_{P_0}[\eta(T) \mid T \in B_s].$$

Let $\eta = \eta^+ - \eta^-$ be the Jordan decomposition of η, where η^+ and η^- are nondecreasing and

$$\|\eta\|_{TV} = \|\eta^+\|_{TV} + \|\eta^-\|_{TV} \le V.$$

Then

$$\eta_n(s) = \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] - \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s].$$

If $s_1 < s_2$, then monotonicity of f_n implies every point of B_{s_1} is no larger than every point of B_{s_2}. Hence the conditional distribution of $T \mid T \in B_{s_2}$ first-order stochastically dominates that of $T \mid T \in B_{s_1}$. Therefore, for any nondecreasing function g,

$$\mathbb{E}_{P_0}[g(T) \mid T \in B_{s_1}] \leq \mathbb{E}_{P_0}[g(T) \mid T \in B_{s_2}].$$

Applying this with $g = \eta^+$ and $g = \eta^-$, both maps

$$s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] \qquad \text{and} \qquad s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]$$

are nondecreasing.

Moreover, each of these conditional expectations takes values in the range of the corresponding function, so

$$\left\| s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] \right\|_{TV} \leq \|\eta^+\|_{TV}, \qquad \left\| s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s] \right\|_{TV} \leq \|\eta^-\|_{TV}.$$

Hence

$$\|\eta_n\|_{TV} \leq \|\eta^+\|_{TV} + \|\eta^-\|_{TV} = \|\eta\|_{TV} \leq V.$$

Finally, if $|\eta(t)| \leq C_0$ on the support of T, then

$$|\eta_n(s)| = |\mathbb{E}_{P_0}[\eta(T) \mid T \in B_s]| \leq C_0$$

for all s in the support of S_n. □

Lemma 3 (Empirical calibration centers the limit)**.** *Suppose Assumptions 1 and 2 hold and the calibration score equations contain the constant score $h \equiv 1$. Then*

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

Proof. Let $T := m(X)$, and write $m_0 = f_0(T)$, where

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \mathbb{E}_{P_0}\left[(Y - f(T))^2\right].$$

Because $\mathcal{F}_{\text{iso}}$ is closed under addition of constants, for every $c \in \mathbb{R}$ the function $f_0 + c$ also belongs to $\mathcal{F}_{\text{iso}}$. Hence the function

$$g(c) := \mathbb{E}_{P_0}\left[(Y - f_0(T) - c)^2\right]$$

is minimized at $c = 0$. Since g is differentiable,

$$0 = g'(0) = -2\,\mathbb{E}_{P_0}[Y - f_0(T)].$$

Therefore

$$\mathbb{E}_{P_0}[Y] = \mathbb{E}_{P_0}[f_0(T)].$$

Recalling that $m_0(X) = f_0(m(X)) = f_0(T)$ and $\psi_0 = \mathbb{E}_{P_0}[Y]$, we conclude

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

□

Proof convention. In the following proofs, fix $\delta > 0$. By $\sup_x |m^\star_{n,\text{iso}}(x)| = O_p(1)$, there exist a deterministic constant $C = C_\delta \geq C_0$ and $n_0 < \infty$ such that the event

$$A_{n,C} := \left\{ \sup_x |m^\star_{n,\text{iso}}(x)| \leq C \right\}$$

satisfies $P_0(A_{n,C}) \geq 1 - \delta$ for all $n \geq n_0$. We carry out the deterministic bounded-class arguments on $A_{n,C}$, where all implicit constants may depend on C but not on n. Since δ is arbitrary, the resulting rates and distributional statements hold unconditionally.

Lemma 4 (Isotonic L^2 rate)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$\|m^\star_{n,\mathrm{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Proof of Lemma 4. Let $T := m(X)$ and let

$$\mathcal{M}_C := \{f(T) : f \in \mathcal{F}_{\mathrm{iso}},\ \|f(T)\|_\infty \le C\}.$$

By Assumption 2, $g_0(T)$ is bounded by $C_0 \le C$, and on $A_{n,C}$ we have $m^\star_{n,\mathrm{iso}} \in \mathcal{M}_C$. Since clipping any monotone candidate to $[-C, C]$ preserves monotonicity and can only decrease squared risk against $g_0(T)$, the population isotonic projection m_0 also belongs to $\mathcal{M}_C$. Thus the problem is a one-dimensional isotonic least-squares regression of Y on the score T. Let

$$g_0(T) := \mathbb{E}_{P_0}[Y \mid T].$$

Then, for every $m \in \mathcal{M}_C$,

$$P_0\{(Y - m)^2\} = P_0\{(Y - g_0)^2\} + P_0\{(g_0 - m)^2\},$$

which shows that m_0 is the $L^2(P_T)$-projection of g_0 onto the closed convex set $\mathcal{M}_C$.

Now define the centered class

$$\mathcal{F}_0 := \{g = m - m_0 :\ m \in \mathcal{M}_C\}.$$

For $g \in \mathcal{F}_0$, let $\ell_g(Z) := (Y - m_0(T) - g(T))^2 - (Y - m_0(T))^2$, where $Z = (X, Y)$. Since $m^\star_{n,\mathrm{iso}}$ minimizes the empirical squared risk over $\mathcal{F}_{\mathrm{iso}}$, on $A_{n,C}$ the random element $\hat{g} := m^\star_{n,\mathrm{iso}} - m_0$ belongs to $\mathcal{F}_0$ and satisfies

$$\mathbb{P}^L_n \ell_{\hat{g}} \le 0 = \mathbb{P}^L_n \ell_0.$$

Moreover,

$$P_0 \ell_g = \|g\|^2_{2,P_{0,X}} - 2P_0\left[(Y - m_0)g\right].$$

Because g is measurable with respect to T, the tower property gives

$$P_0\left[(Y - m_0)g\right] = P_0\left[(g_0 - m_0)g\right].$$

Since m_0 is the $L^2(P_T)$-projection of g_0 onto the convex set $\mathcal{M}_C$, the Hilbert-space projection inequality yields

$$P_0\left[(g_0 - m_0)(m - m_0)\right] \le 0 \qquad \text{for every } m \in \mathcal{M}_C.$$

Hence

$$P_0 \ell_g \ge \|g\|^2_{2,P_{0,X}},$$

so squared loss has the required quadratic margin around m_0.

We next bound the modulus of continuity of the empirical process indexed by the localized classes

$$\mathcal{F}_\delta := \{g \in \mathcal{F}_0 : \|g\|_{2,P_{0,X}} \le \delta\}.$$

Because every $m \in \mathcal{M}_C$ is bounded by C, every $g \in \mathcal{F}_\delta$ is uniformly bounded, and the standard bracketing bound for one-dimensional monotone classes gives

$$\log N_{[]}(\varepsilon, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta/\varepsilon, \qquad 0 < \varepsilon < \delta.$$

Therefore

$$J_{[]}(\delta, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta^{1/2};$$

see, e.g., van der Vaart and Wellner (1996).

Write $\xi := Y - m_0(T)$, so that $\ell_g = g^2 - 2\xi g$. The class $\{g^2 : g \in \mathcal{F}_\delta\}$ is a bounded Lipschitz image of $\mathcal{F}_\delta$, so by Lemma 3.4.2 of van der Vaart and Wellner (1996) and the same entropy bound,

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\,|(\mathbb{P}^L_n - P_0)(g^2)| \lesssim \delta^{1/2}.$$

For the multiplier term, Assumption 2 implies that $Y - g_0(T)$ is sub-Gaussian or subexponential. Since both $g_0(T)$ and $m_0(T)$ are bounded by C, the difference $g_0(T) - m_0(T)$ is bounded, and therefore $\xi = Y - m_0(T)$ is also sub-Gaussian or subexponential. Thus the product class $\xi\mathcal{F}_\delta := \{\xi g : g \in \mathcal{F}_\delta\}$ satisfies the same localized bracketing bound in the Bernstein-type norm used in the subexponential least-squares proof of Bibaut and van der Laan (2019), yielding

$$E^* \sup_{g\in\mathcal{F}_\delta} \sqrt{n}\,|(\mathbb{P}_n^L - P_0)(\xi g)| \lesssim \delta^{1/2}.$$

Combining the previous two displays,

$$E^* \sup_{g\in\mathcal{F}_\delta} \sqrt{n}\,|(\mathbb{P}_n^L - P_0)\ell_g| \lesssim \delta^{1/2}.$$

We may therefore apply Theorem 3.4.1 of van der Vaart and Wellner (1996) with $d(g) = \|g\|_{2,P_{0,X}}$, quadratic margin $P_0\ell_g \gtrsim d^2(g)$, and modulus $\phi_n(\delta) \asymp \delta^{1/2}$. The fixed-point condition $r_n^2\phi_n(r_n^{-1}) \lesssim \sqrt{n}$ gives $r_n \asymp n^{1/3}$, and hence

$$\|m^\star_{n,\mathrm{iso}} - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|m^\star_{n,\mathrm{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

The preceding display is obtained on $A_{n,C}$. Since $P_0(A^c_{n,C}) \le \delta$ for all large n, and $\delta > 0$ was arbitrary, the same rate holds unconditionally. This is the classical one-dimensional isotonic rate, stated directly in the random-design $L^2(P_{0,X})$ norm relevant for the theorem. □

Theorem 4 (Calibration error of isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{m}_0(X) := \mathbb{E}_{P_0}[Y \mid m^\star_{n,\mathrm{iso}}(X)].$$

Then

$$\|\widehat{m}_0 - m^\star_{n,\mathrm{iso}}\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Proof. The proof follows that of Theorem 1 in van der Laan et al. (2023b); see also van der Laan and Alaa (2025). Let

$$T := m(X), \qquad S_n := m^\star_{n,\mathrm{iso}}(X), \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s],$$

and define

$$\Delta_n(X) := \widehat{m}_0(X) - m^\star_{n,\mathrm{iso}}(X) = \eta_n(S_n) - S_n.$$

Conditional on the labeled sample, $h_n(s) := \eta_n(s) - s$ is a measurable function of s. Applying proposition 2 pathwise with $h = h_n$ yields

$$\mathbb{P}_n^L[\Delta_n(X)\{Y - S_n\}] = 0.$$

Also,

$$P_0[\Delta_n(X)\{Y - S_n\}] = P_0[\Delta_n(X)\,\mathbb{E}_{P_0}[Y - S_n \mid S_n]] = P_0(\Delta_n^2) = \|\Delta_n\|^2_{2,P_{0,X}}.$$

Hence

$$\|\Delta_n\|^2_{2,P_{0,X}} = (P_0 - \mathbb{P}_n^L)[\Delta_n(X)\{Y - S_n\}]\,.$$

Fix $\delta > 0$, and work on the event $A_{n,C}$ from the proof convention. On this event, S_n is uniformly bounded by C. By Lemma 2, the function η_n belongs to the bounded-variation class

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$

where $V < \infty$ is the variation bound in Assumption 3. Let

$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\}.$$

Since $S_n = f_n(T)$ for a nondecreasing fitted isotonic map f_n, and composition with a monotone map preserves bounded variation, the function

$$t \mapsto \eta_n\{f_n(t)\}$$

also belongs to $\mathcal{F}_{\mathrm{TV},C}$, after enlarging C if necessary. Thus $\eta_n(S_n) = h_1(T)$ for some $h_1 = f_1 \circ T$ with $f_1 \in \mathcal{F}_{\mathrm{TV},C}$, while $S_n = h_2(T)$ for $h_2 = f_2 \circ T$ with $f_2 = f_n \in \mathcal{F}_{\mathrm{iso},C}$.

Define the localized product class

$$\mathcal{G}_{n,C,\delta} := \Big\{(h_1 - h_2)(Y - h_2) : h_1 = f_1 \circ T,\ h_2 = f_2 \circ T,\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C},\ \|h_1 - h_2\|_{2,P_{0,X}} \le \delta\Big\}.$$

Then

$$\Delta_n(X)\{Y - S_n\} \in \mathcal{G}_{n,C,\delta_n}, \qquad \delta_n := \|\Delta_n\|_{2,P_{0,X}}.$$

Because bounded-variation functions are differences of bounded monotone functions, standard entropy-preservation results for Lipschitz transformations imply that the localized entropy integral of $\mathcal{G}_{n,C,\delta}$ satisfies

$$J_{[]}(\delta, \mathcal{G}_{n,C,\delta}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Moreover, Assumption 2 implies that $Y - \eta(T)$ is sub-Gaussian or subexponential, and on $A_{n,C}$ the difference $\eta(T) - S_n$ is bounded. Therefore $Y - S_n$ has the same tail type, up to constants. The same localized multiplier empirical-process argument used in the proof of Lemma 4 then gives

$$E^* \sup_{g \in \mathcal{G}_{n,C,\delta}} |(\mathbb{P}_n^L - P_0)g| \lesssim n^{-1/2}\delta^{1/2}.$$

Set $\varepsilon_n := n^{-1/3}$. For $s \ge 0$, let

$$B_s := \{2^s\varepsilon_n \le \delta_n < 2^{s+1}\varepsilon_n\}.$$

On $B_s \cap A_{n,C}$,

$$2^{2s}\varepsilon_n^2 \le \delta_n^2 \le \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)g|.$$

Therefore, by Markov's inequality,

$$P_0(B_s \cap A_{n,C}) \lesssim \frac{E^* \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)g|}{2^{2s}\varepsilon_n^2} \lesssim \frac{n^{-1/2}(2^{s+1}\varepsilon_n)^{1/2}}{2^{2s}\varepsilon_n^2} \lesssim 2^{-3s/2},$$

since $n^{-1/2}\varepsilon_n^{-3/2} = 1$. Summing over $s \ge S$ yields

$$P_0(\delta_n \ge 2^S\varepsilon_n,\ A_{n,C}) \lesssim \sum_{s=S}^{\infty} 2^{-3s/2} \xrightarrow[S\to\infty]{} 0.$$

Because $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ is arbitrary, it follows that

$$\|\Delta_n\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

This is exactly the claimed calibration-error rate. □

Proof of Lemma 3. Because $h \equiv 1$ belongs to the calibration score class, equation (8) gives

$$\mathbb{P}_n^L\{Y - m_{n,\mathrm{iso}}^\star(X)\} = 0.$$

Hence

$$\mathbb{P}_n^L Y - \mathbb{P}_n^L m_0(X) = \mathbb{P}_n^L\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\}.$$

By Cauchy–Schwarz and Lemma 4,

$$P_{0,X}\big|m_{n,\mathrm{iso}}^\star - m_0\big| \le \|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}),$$

and Lemma 5 gives

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p(n^{-2/3}).$$

Therefore

$$\mathbb{P}_n^L\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\} = P_{0,X}\{m_{n,\mathrm{iso}}^\star - m_0\} + (\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = o_p(1).$$

By Assumption 1, the law of large numbers yields $\mathbb{P}_n^L Y \to_p \psi_0$ and $\mathbb{P}_n^L m_0(X) \to_p \mathbb{E}_{P_{0,X}}[m_0(X)]$. Therefore $\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0$. □

Lemma 5 (Centered empirical-process bounds for isotonic calibration)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\} = O_p(n^{-2/3}), \qquad (\mathbb{P}_N^U - P_{0,X})\{m_{n,\mathrm{iso}}^\star(\widetilde{X}) - m_0(\widetilde{X})\} = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof of Lemma 5. By Lemma 4,

$$\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

To control the centered empirical-process term, work on the event $A_{n,C}$ from the proof convention and consider the localized class

$$\mathcal{F}_{n,C} := \{g = f \circ m - m_0 : \ f \in \mathcal{F}_{\mathrm{iso}}, \ \|f \circ m\|_\infty \le C, \ \|g\|_{2,P_{0,X}} \lesssim n^{-1/3}\}.$$

On $A_{n,C}$, the function $m_{n,\mathrm{iso}}^\star - m_0$ belongs to $\mathcal{F}_{n,C}$. Because m is fixed, this is a localized class of bounded monotone functions of a one-dimensional score. Standard bracketing entropy bounds for monotone classes imply

$$J_{[]}(r, \mathcal{F}_{n,C}, L_2(P_0)) \lesssim r^{1/2}.$$

Applying a local maximal inequality for empirical processes, such as Lemma 3.4.2 of van der Vaart and Wellner (1996), at radius $r_n \asymp n^{-1/3}$ yields

$$\sup_{g \in \mathcal{F}_{n,C}} |(\mathbb{P}_n^L - P_{0,X})g| = O_p\Big(n^{-1/2} r_n^{1/2}\Big) = O_p(n^{-2/3}).$$

In particular,

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ is arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, $m_{n,\mathrm{iso}}^\star - m_0$ is fixed and the unlabeled sample is independent, so

$$(\mathbb{P}_N^U - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p\left(N^{-1/2}\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by Lemma 4. □

Lemma 6 (Centered empirical-process bounds for isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\Delta_n(X) := \widehat{m}_0(X) - m_{n,\mathrm{iso}}^\star(X).$$

Then

$$(\mathbb{P}_n^L - P_{0,X})\Delta_n(X) = O_p(n^{-2/3}), \qquad (\mathbb{P}_N^U - P_{0,X})\Delta_n(\widetilde{X}) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof. By theorem 4,

$$\|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Let $V < \infty$ be the variation bound in Assumption 3. Fix $\delta > 0$, and work on the event $A_{n,C}$. On this event, $m_{n,\mathrm{iso}}^\star$ is uniformly bounded by C. By Lemma 2, the function

$$s \mapsto \eta_n(s) := \mathbb{E}_{P_0}[Y \mid m_{n,\mathrm{iso}}^\star(X) = s]$$

has total variation at most V and is bounded by C on the support of $m_{n,\mathrm{iso}}^\star(X)$. Since $m_{n,\mathrm{iso}}^\star(X) = f_n(T)$ with $T = m(X)$ and f_n nondecreasing, it follows that

$$\widehat{m}_0(X) = \eta_n\{f_n(T)\}$$

is a bounded-variation transform of T, while $m_{n,\mathrm{iso}}^\star(X)$ is a bounded monotone transform of T.

Consider the classes

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$
$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\},$$

and define

$$\mathcal{H}_{n,C} := \Big\{h_1 - h_2 : h_1 = f_1 \circ T,\ h_2 = f_2 \circ T,\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C},\ \|h_1 - h_2\|^2_{2,P_{0,X}} \lesssim n^{-2/3}\Big\}.$$

On $A_{n,C}$, the random function Δ_n belongs to $\mathcal{H}_{n,C}$.

Because bounded-variation functions are differences of bounded monotone functions, the localized bracketing entropy of $\mathcal{H}_{n,C}$ obeys the same bound as in the proof of theorem 4, namely

$$J_{[]}(\delta, \mathcal{H}_{n,C}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Applying the same local maximal inequality at localization radius $n^{-1/3}$ therefore gives, on $A_{n,C}$,

$$|(\mathbb{P}_n^L - P_{0,X})\Delta_n(X)| = O_p(n^{-2/3}).$$

Since $P_0(A^c_{n,C}) \le \delta$ for all large n, and $\delta > 0$ was arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, Δ_n is fixed and the unlabeled sample is independent. Hence

$$(\mathbb{P}_N^U - P_{0,X})\Delta_n(\widetilde{X}) = O_p\left(N^{-1/2}\|\Delta_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by theorem 4. □

I Proof of asymptotic linearity

Proof of theorem 2. The proof combines the exact AIPW representation from Theorem 1 with a localized empirical-process bound for the isotonic calibrator class. The latter controls the remainder generated by estimating the monotone post-processing map and yields the $n^{-2/3}$ remainder rate. The overall argument is closely related to the calibrated DML framework of van der Laan et al. (2024c); see also Kennedy (2024) for background on semiparametric influence-function arguments.

By Lemma 3, $P_{0,X}m_0 = \psi_0$. Using theorem 1 with $\widehat{w} \equiv 1$,

$$\widehat{\psi}_{\mathrm{iso}} = \rho_n\mathbb{P}_n^L\{m^\star_{n,\mathrm{iso}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\mathrm{iso}}(\widetilde{X})\} + \rho_n\mathbb{P}_n^L\{Y - m^\star_{n,\mathrm{iso}}(X)\}.$$

Add and subtract the AIPW form built from m_0:

$$\widehat{\psi}_{\mathrm{iso}} = \rho_n\mathbb{P}_n^L\left\{m_0 + \rho_n^{-1}(Y - m_0)\right\} + (1-\rho_n)\mathbb{P}_N^U\{m_0(\widetilde{X})\} + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m^\star_{n,\mathrm{iso}} - m_0\}.$$

Subtracting ψ_0 gives

$$\begin{aligned}\widehat{\psi}_{\mathrm{iso}} - \psi_0 = \rho_n(\mathbb{P}_n^L - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m_0 - \psi_0\} \\ + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m^\star_{n,\mathrm{iso}} - m_0\}.\end{aligned}$$

To replace ρ_n^{-1} by ρ_0^{-1}, note that

$$\rho_n(\mathbb{P}_n^L - P_0)\Big\{(\rho_n^{-1} - \rho_0^{-1})(Y - m_0)\Big\} = \Big(1 - \frac{\rho_n}{\rho_0}\Big)(\mathbb{P}_n^L - P_0)(Y - m_0).$$

The right-hand side is $o_p(M^{-1/2})$ because $\rho_n \to \rho_0$ and

$$(\mathbb{P}_n^L - P_0)(Y - m_0) = O_p(n^{-1/2}) = O_p(M^{-1/2}).$$

Also,

$$(1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\text{iso}}^\star - m_0\} = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m_{n,\text{iso}}^\star - m_0\} - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\{m_{n,\text{iso}}^\star - m_0\}.$$

By Lemma 5, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Thus the remainder $R_{n,N}$, defined as the difference between $\widehat{\psi}_{\text{iso}} - \psi_0$ and the leading empirical-process terms in equation (18), satisfies the bound stated in theorem 2. Since $n^{-2/3} = o(M^{-1/2})$ and $n^{-1/3}N^{-1/2} = o(M^{-1/2})$, this yields equation (18). □

Proof of corollary 1. By theorem 2,

$$\sqrt{M}\,(\widehat{\psi}_{\text{iso}} - \psi_0) = \frac{1}{\sqrt{M}}\sum_{i=1}^{n} D_{m_0}^L(X_i, Y_i) + \frac{1}{\sqrt{M}}\sum_{j=1}^{N} D_{m_0}^U(\widetilde{X}_j) + o_p(1).$$

The labeled and unlabeled samples are independent, each summand has mean zero, and $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1. A standard triangular-array central limit theorem therefore yields the stated asymptotic normal law with variance σ_0^2.

For the variance estimator, let $g_n := m_{n,\text{iso}}^\star - m_0$. By Lemma 4,

$$P_{0,X}(g_n^2) = \|g_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Moreover, the proof of Lemma 4 already established the localized empirical-process bound

$$\sup_{g\in\mathcal{F}_\delta} |(\mathbb{P}_n^L - P_{0,X})(g^2)| = O_p(n^{-1/2}\delta^{1/2}),$$

for the squared class $\{g^2 : g \in \mathcal{F}_\delta\}$. Taking $\delta \asymp n^{-1/3}$, which matches the localization radius of g_n, gives

$$(\mathbb{P}_n^L - P_{0,X})(g_n^2) = O_p(n^{-2/3}).$$

Hence

$$\mathbb{P}_n^L(g_n^2) = O_p(n^{-2/3}) = o_p(1).$$

On the boundedness event $A_{n,C}$, conditional on the labeled data, g_n is fixed and $|g_n| \le 2C$, so

$$(\mathbb{P}_N^U - P_{0,X})(g_n^2) = O_p\left(N^{-1/2}\|g_n^2\|_{2,P_{0,X}}\right) = O_p\left(C\,N^{-1/2}\|g_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Therefore

$$\mathbb{P}_N^U(g_n^2) = O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big) = o_p(1).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n and arbitrary $\delta > 0$, these bounds hold unconditionally. Also, $\widehat{\psi}_{\text{iso}} \to_p \psi_0$ follows from theorem 2.

Finally,

$$\widehat{D}^L - D_{m_0}^L = (1-\rho_n^{-1})g_n - (\widehat{\psi}_{\text{iso}} - \psi_0) + (\rho_n^{-1} - \rho_0^{-1})(Y - m_0),$$

and

$$\widehat{D}^U - D_{m_0}^U = g_n(\widetilde{X}) - (\widehat{\psi}_{\text{iso}} - \psi_0).$$

Because $\rho_n \to \rho_0$, $Y - m_0$ has finite second moment, and $\mathbb{P}_n^L(g_n^2), \mathbb{P}_N^U(g_n^2) = o_p(1)$, it follows that

$$\mathbb{P}_n^L\Big[\{\widehat{D}^L - D_{m_0}^L\}^2\Big] = o_p(1), \qquad \mathbb{P}_N^U\Big[\{\widehat{D}^U - D_{m_0}^U\}^2\Big] = o_p(1).$$

Therefore $\widehat{\sigma}^2 \to_p \sigma_0^2$. The Wald interval claim then follows by Slutsky's theorem. □

J Proof of post-processing equivalence and reduced-model efficiency

Proof of proposition 5. Let

$$\Delta_n(X) := \widehat{m}_0(X) - m^\star_{n,\mathrm{iso}}(X).$$

Using the definition of the AIPW class together with $\mathbb{P}_n^L\{Y - m^\star_{n,\mathrm{iso}}(X)\} = 0$, we have

$$\begin{aligned}\widehat{\psi}_{\mathrm{eff}} - \widehat{\psi}_{\mathrm{iso}} &= \rho_n \mathbb{P}_n^L \Delta_n(X) + (1-\rho_n)\mathbb{P}_N^U \Delta_n(\widetilde{X}) + \mathbb{P}_n^L\{Y - \widehat{m}_0(X)\} - \mathbb{P}_n^L\{Y - m^\star_{n,\mathrm{iso}}(X)\} \\ &= \rho_n \mathbb{P}_n^L \Delta_n(X) + (1-\rho_n)\mathbb{P}_N^U \Delta_n(\widetilde{X}) - \mathbb{P}_n^L \Delta_n(X) \\ &= (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n.\end{aligned}$$

Now decompose

$$(1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\Delta_n - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\Delta_n.$$

By Lemma 6, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Since $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1, both rates are $o_p(M^{-1/2})$. Therefore

$$\widehat{\psi}_{\mathrm{eff}} - \widehat{\psi}_{\mathrm{iso}} = (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n = o_p(M^{-1/2}).$$

This proves the claimed first-order equivalence. In particular, $\widehat{\psi}_{\mathrm{eff}}$ and $\widehat{\psi}_{\mathrm{iso}}$ have the same asymptotic variance. □

Proof of corollary 2. For part (i), if $m_0 = \mu_0$ almost surely, then the asymptotic influence pair in theorem 2 is

$$D^L_{\mu_0}(X,Y) = \mu_0(X) - \psi_0 + \rho_0^{-1}\{Y - \mu_0(X)\}, \qquad D^U_{\mu_0}(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0.$$

By proposition 1, this is the efficient influence pair in the unrestricted two-sample model. Therefore $\widehat{\psi}_{\mathrm{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

For part (ii), let

$$S := m(X), \qquad \eta_0(s) := \mathbb{E}_{P_0}[Y \mid S = s].$$

proposition 1 applied to the reduced two-sample experiment with scalar covariate S shows that the efficient influence pair in that reduced experiment is

$$D^L_{\eta_0}(S,Y) = \eta_0(S) - \psi_0 + \rho_0^{-1}\{Y - \eta_0(S)\}, \qquad D^U_{\eta_0}(\widetilde{S}) = \eta_0(\widetilde{S}) - \psi_0.$$

If η_0 is monotone increasing, then η_0 belongs to the isotonic class and

$$\mathbb{E}_{P_0}\left[\{Y - g(S)\}^2\right] = \mathbb{E}_{P_0}\left[\{Y - \eta_0(S)\}^2\right] + \mathbb{E}_{P_0}\left[\{\eta_0(S) - g(S)\}^2\right]$$

for every measurable g. Hence the population isotonic regression of Y on the score $S = m(X)$ is exactly $\eta_0(S)$, so

$$m_0(X) = \eta_0\{m(X)\} \qquad \text{almost surely.}$$

Therefore, the influence pair in theorem 2 coincides with the reduced-model efficient influence pair above, so $\widehat{\psi}_{\mathrm{iso}}$ is efficient for ψ_0 in the reduced experiment indexed by $m(X)$. □

K Pooled i.i.d. Missing-Data Formulation

The joint two-sample setup can be embedded into a pooled i.i.d. model by introducing a sample-membership indicator $R \in \{0,1\}$, where $R = 1$ denotes labeled observations and $R = 0$ denotes unlabeled observations. In that formulation one observes i.i.d. data $O = (X, R, RY)$ with $\mathbb{P}(R = 1) = \rho_0$, and the calibrated estimator becomes

$$\widehat{\psi}_{\mathrm{cal}} = \mathbb{P}_M(m_n^\star) + \mathbb{P}_M\left[\frac{R}{\rho_0}\{Y - m_n^\star\}\right]$$

up to the same calibration identity as in the main text. The corresponding observed-data efficient influence function is

$$m_0(X) - \psi_0 + \frac{R}{\rho_0}\{Y - m_0(X)\},$$

which is the single-sample analogue of the efficient influence pair obtained by applying proposition 1 to the reduced model with scalar covariate $m(X)$. Thus the usual i.i.d. missing-data formulation is a convenient special case, but the main text keeps the two-dataset structure explicit.

L Additional empirical details

This appendix records the benchmark construction used in section 5.2. We use the official `ppi_py` real-data mean-estimation examples and preserve their labeled-sample-size grids. The `forest` experiment uses $n \in \{50, 100, \ldots, 500\}$. The `galaxies` and semisupervised `census_income` benchmarks use $n \in \{50, 155, 261, 366, 472, 577, 683, 788, 894, 1000\}$. For each n, the remaining observations form the unlabeled sample. In the current draft we rerandomize this split 500 times at each n, always evaluating all estimators on the same split and taking the full-sample mean as the benchmark truth.

The estimator set mirrors the original PPI comparison wherever possible. In the paper-facing summaries, we report the labeled-only estimator, the imputation benchmark on binary-outcome tasks, PPI computed using the official `ppi_py` implementation, the classical AIPW estimator based on the same prediction score, clipped PPI++, and the unclipped efficiency-maximized benchmark AIPW-EM. We then add five calibration-oriented comparators: linear calibration; smooth monotone spline calibration; Platt scaling on the binary-outcome benchmarks; isotonic calibration with a fixed minimum bin size of 10; and an adaptive `AutoCal` selector that chooses among AIPW, linear calibration, monotone spline calibration, and isotonic calibration by cross-validated empirical efficiency. For the binary-outcome benchmarks, we also report Venn–Abers shrinkage. Platt scaling and Venn–Abers are omitted on `census_income` because that outcome is not binary. Including both PPI++ and AIPW-EM lets us separate the underlying efficiency-maximization idea from the specific `ppi_py` clipping rule for the tuning coefficient, while figure 3 isolates the fixed-calibration comparison among PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`.

For the fixed calibration-based plug-in estimators, we fit the calibrator on the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^n$, transform both the labeled and unlabeled scores, and then compute the pooled plug-in mean together with the same semisupervised influence-style Wald interval used throughout the benchmark code. AIPW uses the same score but averages it over the pooled covariate sample before adding the labeled residual correction, whereas PPI uses only the unlabeled score average. In theory this retains less labeled-sample score information, though in the finite-sample benchmark results the empirical variance gap is often small and not uniform across all datasets. PPI++ and AIPW-EM both apply one-dimensional empirical efficiency maximization to the same score, but PPI++ inherits the official `ppi_py` clipping rule $\lambda \in [0, 1]$ whereas AIPW-EM does not. Linear calibration regresses Y on an intercept and the score, then clips the fitted values to the range of predicted outcomes observed in the labeled data to prevent extrapolation on the unlabeled data. `MonoSpline` instead fits a smooth nondecreasing spline of Y on the score, using monotonicity constraints together with a mild roughness penalty, and then applies the same clipped pooled plug-in construction. Platt scaling fits a logistic regression of the binary outcome on a stabilized logit transform of the score, while the fixed-bin isotonic estimator fits a stepwise monotone calibration map and also clips to the labeled outcome range. The `AutoCal` estimator is a selector rather than a new calibrator: in the experiments it compares AIPW, linear calibration, monotone spline calibration, and fixed-bin isotonic calibration using 20-fold cross-validation and the estimated influence-function variance as the selection criterion. To keep this step inexpensive when the unlabeled sample is very large, the foldwise criterion is evaluated on an unlabeled subsample of size $\min(N, 10n)$. The reported summaries are Monte Carlo averages of bias, empirical variance, MSE, interval coverage, and relative efficiency versus PPI across the repeated random splits. The complete numerical outputs, including dataset-level tables and per-metric plots, are generated automatically by the accompanying reproduction pipeline.

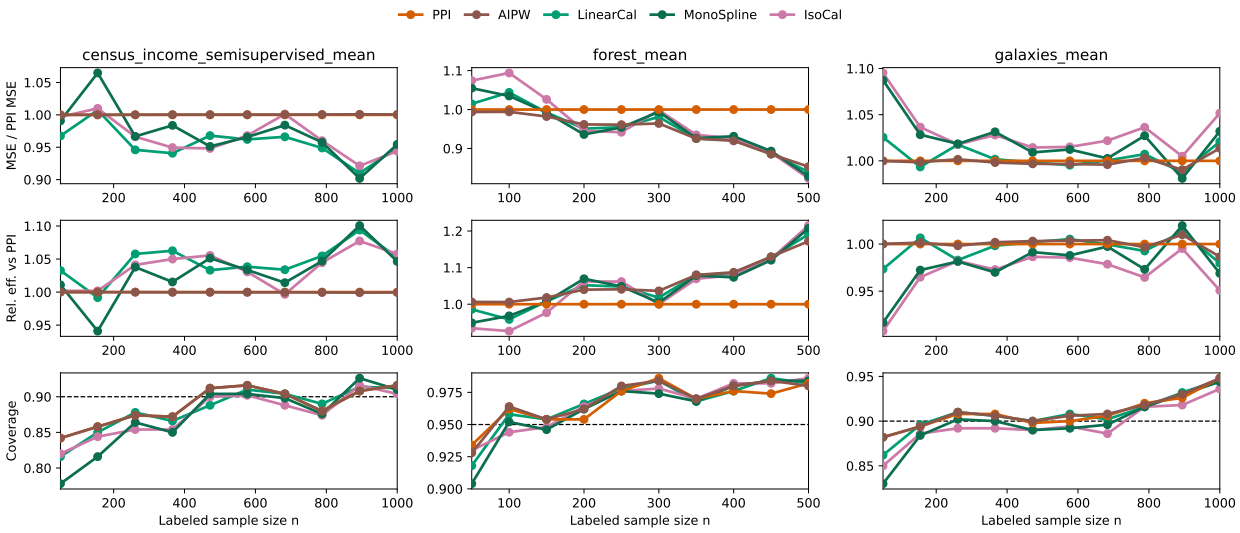

Figure 3: Calibration-focused appendix comparison across the reproduced PPI benchmarks, restricted to PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`. The panels report normalized MSE relative to PPI, relative efficiency versus PPI, and coverage across labeled-sample-size regimes. This figure isolates the fixed-calibration comparison that is omitted from the main-text benchmark figure for readability.

M Python Code

Code availability. A public repository containing the `ppi_aipw` package, experiment scripts, and paper assets is available at github.com/Larsvanderlaan/ppi-aipw. Package documentation and worked examples are hosted at larsvanderlaan.github.io/ppi-aipw/.

Below, we provide a minimal, self-contained implementation of the method introduced in this paper.

M.1 Isotonic regression via XGBoost

```
import numpy as np
import xgboost as xgb

def isoreg_with_xgboost(x, y, max_depth=15, min_child_weight=20, weights=None):
    """
    Fit a monotone calibrator f so that f(x) is nondecreasing in x.

    Parameters
    ----------
    x : array-like, shape (n,) or (n, 1)
        Predictor used for calibration. In the PPI application this is usually
        the score m(X), so x is typically one-dimensional.
    y : array-like, shape (n,)
        Labeled outcomes.
    max_depth : int, default=15
        Maximum depth for the one-round XGBoost fit.
    min_child_weight : float, default=20
        Minimum total weight in a terminal node.
    weights : array-like, optional
        Optional observation weights.

    Returns
```

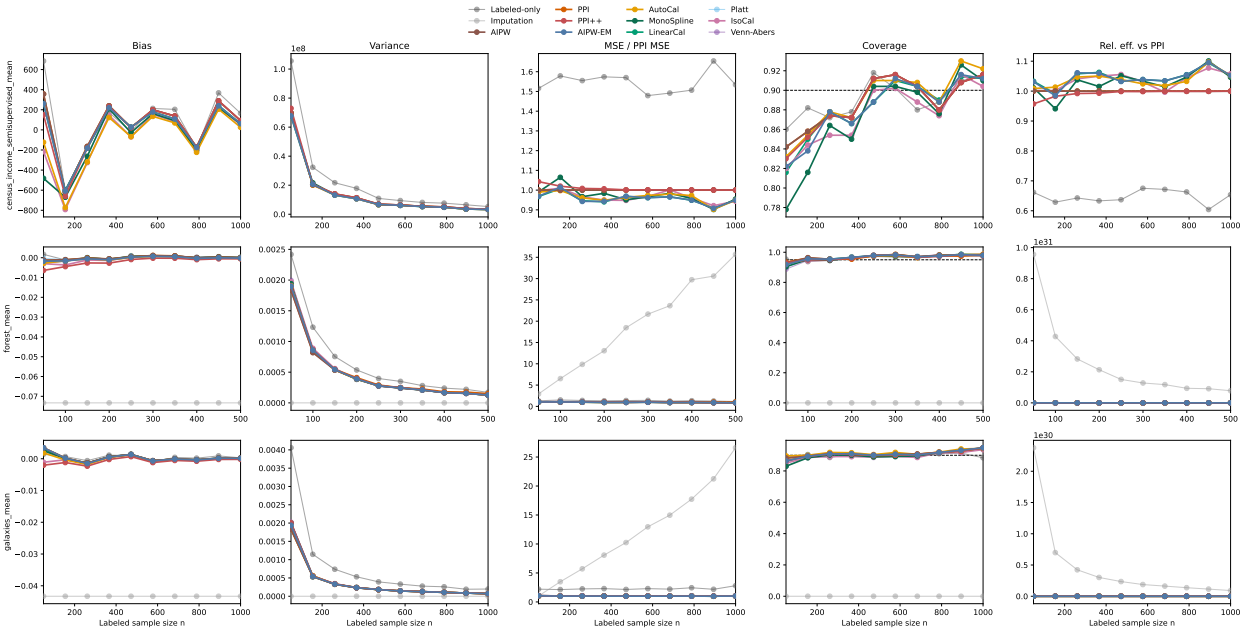

Figure 4: Full diagnostic grid for the reproduced PPI benchmarks, showing bias, empirical variance, normalized MSE relative to PPI, coverage, and relative efficiency versus PPI across labeled-sample-size regimes. In the normalized-MSE and relative-efficiency panels, the dashed horizontal line marks parity with PPI; in the coverage panels it marks the nominal target coverage $1 - \alpha$. `AutoCal`, `MonoSpline`, and `IsoCal` appear alongside PPI, AIPW, PPI++, AIPW-EM, and the fixed calibration rules, while Platt scaling and Venn–Abers appear only on the binary-outcome datasets where they are well defined.

```
-------
predict_fn : callable
    Function mapping new x values to calibrated predictions f(x).
"""
x = np.asarray(x)
y = np.asarray(y).reshape(-1)
x = x.reshape(len(y), -1)

data = xgb.DMatrix(data=x, label=y, weight=weights)
params = {
    "max_depth": max_depth,
    "min_child_weight": min_child_weight,
    "monotone_constraints": "(" + ",".join(["1"] * x.shape[1]) + ")",
    "eta": 1.0,
    "gamma": 0.0,
    "lambda": 0.0,
    "objective": "reg:squarederror",
    "verbosity": 0,
}
iso_fit = xgb.train(params=params, dtrain=data, num_boost_round=1)

def predict_fn(x_new):
    x_new = np.asarray(x_new)
    if x_new.ndim == 1:
        x_new = x_new.reshape(-1, 1)
    data_pred = xgb.DMatrix(data=x_new)
    return iso_fit.predict(data_pred)
```

```
    return predict_fn
```

M.2 Isotonic-calibrated plug-in

```
import numpy as np
from statistics import NormalDist


def _resolve_scores(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    Y_labeled = np.asarray(Y_labeled).reshape(-1)

    if score_fn is not None:
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn together with X_labeled/X_unlabeled, "
            "or provide score_labeled and score_unlabeled."
        )

    score_labeled = np.asarray(score_labeled).reshape(-1)
    score_unlabeled = np.asarray(score_unlabeled).reshape(-1)

    if len(score_labeled) != len(Y_labeled):
        raise ValueError("Y_labeled and score_labeled must have the same length.")

    return Y_labeled, score_labeled, score_unlabeled


def isotonic_calibrated_plugin(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
    alpha=0.05,
    min_obs=20,
):
    Y_labeled, score_labeled, score_unlabeled = _resolve_scores(
        Y_labeled=Y_labeled,
        X_labeled=X_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
    )
```

```
    n = len(Y_labeled)
    N = len(score_unlabeled)
    M = n + N
    rho = n / M

    calibrator = isoreg_with_xgboost(
        x=score_labeled,
        y=Y_labeled,
        min_child_weight=min_obs,
    )

    m_labeled = calibrator(score_labeled)
    m_unlabeled = calibrator(score_unlabeled)

    psi_hat = (n * m_labeled.mean() + N * m_unlabeled.mean()) / M

    if_labeled = m_labeled - psi_hat + (Y_labeled - m_labeled) / rho
    if_unlabeled = m_unlabeled - psi_hat
    var_hat = (
        rho * np.var(if_labeled, ddof=1)
        + (1 - rho) * np.var(if_unlabeled, ddof=1)
    )
    se_hat = np.sqrt(var_hat / M)

    z = NormalDist().inv_cdf(1 - alpha / 2)
    ci = (psi_hat - z * se_hat, psi_hat + z * se_hat)

    return {
        "estimate": psi_hat,
        "standard_error": se_hat,
        "confidence_interval": ci,
    }


# Example:
# fit = isotonic_calibrated_plugin(
#     Y_labeled=Y,
#     X_labeled=X,
#     X_unlabeled=X_tilde,
#     score_fn=m,
# )
```

M.3 Linear calibration

```
import numpy as np
from statistics import NormalDist


def _resolve_scores(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    Y_labeled = np.asarray(Y_labeled).reshape(-1)
```

```
    if score_fn is not None:
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn together with X_labeled/X_unlabeled, "
            "or provide score_labeled and score_unlabeled."
        )

    score_labeled = np.asarray(score_labeled).reshape(-1)
    score_unlabeled = np.asarray(score_unlabeled).reshape(-1)

    if len(score_labeled) != len(Y_labeled):
        raise ValueError("Y_labeled and score_labeled must have the same length.")

    return Y_labeled, score_labeled, score_unlabeled


def linear_calibrated_plugin(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
    alpha=0.05,
):
    Y_labeled, score_labeled, score_unlabeled = _resolve_scores(
        Y_labeled=Y_labeled,
        X_labeled=X_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
    )

    n = len(Y_labeled)
    N = len(score_unlabeled)
    M = n + N
    rho = n / M

    X_design = np.column_stack([np.ones(n), score_labeled])
    beta_hat, _, _, _ = np.linalg.lstsq(X_design, Y_labeled, rcond=None)

    def calibrator(score):
        score = np.asarray(score).reshape(-1)
        return beta_hat[0] + beta_hat[1] * score

    m_labeled = calibrator(score_labeled)
    m_unlabeled = calibrator(score_unlabeled)

    psi_hat = (n * m_labeled.mean() + N * m_unlabeled.mean()) / M

    if_labeled = m_labeled - psi_hat + (Y_labeled - m_labeled) / rho
    if_unlabeled = m_unlabeled - psi_hat
```

```
    var_hat = (
        rho * np.var(if_labeled, ddof=1)
        + (1 - rho) * np.var(if_unlabeled, ddof=1)
    )
    se_hat = np.sqrt(var_hat / M)

    z = NormalDist().inv_cdf(1 - alpha / 2)
    ci = (psi_hat - z * se_hat, psi_hat + z * se_hat)

    return {
        "estimate": psi_hat,
        "standard_error": se_hat,
        "confidence_interval": ci,
    }


# Example:
# fit = linear_calibrated_plugin(
#     Y_labeled=Y,
#     score_labeled=mX,
#     score_unlabeled=mX_tilde,
# )
```

Dataset	n	N	Estimator	Bias	Variance	MSE	Coverage	RelEff
census_income_semisupervised_mean	50	380041	AIPW	356.4905	69836665.4410	69963750.9366	0.842	1.000
census_income_semisupervised_mean	50	380041	AIPW-EM	260.9833	67728426.6042	67796538.8913	0.822	1.031
census_income_semisupervised_mean	50	380041	AutoCal	-123.2501	69242944.1124	69258134.7078	0.832	1.009
census_income_semisupervised_mean	50	380041	Labeled-only	685.3532	105558402.2427	106028111.3137	0.860	0.662
census_income_semisupervised_mean	50	380041	IsoCal	-215.3832	69697400.2434	69743790.1536	0.820	1.002
census_income_semisupervised_mean	50	380041	LinearCal	260.9294	67622601.4006	67690685.5430	0.816	1.033
census_income_semisupervised_mean	50	380041	MonoSpline	-482.0477	69082916.1270	69315286.1508	0.778	1.011
census_income_semisupervised_mean	50	380041	PPI	356.4473	69834865.9486	69961920.5977	0.842	1.000
census_income_semisupervised_mean	50	380041	PPI++	150.8664	72953053.8328	72975814.4980	0.830	0.957
census_income_semisupervised_mean	577	379514	AIPW	199.8136	6353884.5833	6393810.0648	0.916	1.000
census_income_semisupervised_mean	577	379514	AIPW-EM	177.6067	6114918.2974	6146462.4536	0.912	1.039
census_income_semisupervised_mean	577	379514	AutoCal	134.3746	6201029.6459	6219086.1828	0.910	1.024
census_income_semisupervised_mean	577	379514	Labeled-only	213.2465	9407595.6062	9453069.6946	0.902	0.675
census_income_semisupervised_mean	577	379514	IsoCal	155.0503	6162964.7883	6187005.3977	0.902	1.031
census_income_semisupervised_mean	577	379514	LinearCal	179.8005	6116832.9552	6149161.1594	0.910	1.038
census_income_semisupervised_mean	577	379514	MonoSpline	162.1952	6146069.6864	6172376.9541	0.904	1.033
census_income_semisupervised_mean	577	379514	PPI	199.7932	6351836.6166	6391753.9369	0.916	1.000
census_income_semisupervised_mean	577	379514	PPI++	198.7409	6362047.6345	6401545.5819	0.916	0.998
census_income_semisupervised_mean	1000	379091	AIPW	92.1091	3462051.4638	3470535.5425	0.916	0.999
census_income_semisupervised_mean	1000	379091	AIPW-EM	61.4705	3288235.4312	3292014.0588	0.912	1.052
census_income_semisupervised_mean	1000	379091	AutoCal	25.8207	3289467.9320	3290134.6381	0.922	1.052
census_income_semisupervised_mean	1000	379091	Labeled-only	165.4763	5293296.2822	5320678.6872	0.914	0.654
census_income_semisupervised_mean	1000	379091	IsoCal	51.2697	3271622.4413	3274251.0223	0.904	1.058
census_income_semisupervised_mean	1000	379091	LinearCal	60.7501	3287796.5296	3291487.1058	0.912	1.052
census_income_semisupervised_mean	1000	379091	MonoSpline	60.9626	3306575.6859	3310292.1289	0.910	1.046
census_income_semisupervised_mean	1000	379091	PPI	91.9155	3460071.7259	3468520.1894	0.916	1.000
census_income_semisupervised_mean	1000	379091	PPI++	91.9155	3460071.7259	3468520.1894	0.916	1.000
forest_mean	50	1546	AIPW	-0.0012	0.0018	0.0018	0.928	1.006
forest_mean	50	1546	AIPW-EM	-0.0018	0.0019	0.0019	0.918	0.969
forest_mean	50	1546	AutoCal	-0.0028	0.0019	0.0019	0.924	0.965
forest_mean	50	1546	Labeled-only	0.0017	0.0024	0.0024	0.954	0.762
forest_mean	50	1546	Imputation	-0.0733	0.0000	0.0054	0.000	9564441057429144180180373209088.000
forest_mean	50	1546	IsoCal	-0.0030	0.0020	0.0020	0.930	0.934
forest_mean	50	1546	LinearCal	-0.0016	0.0019	0.0019	0.918	0.986
forest_mean	50	1546	MonoSpline	-0.0021	0.0019	0.0019	0.904	0.949
forest_mean	50	1546	Platt	-0.0003	0.0019	0.0019	0.900	0.948
forest_mean	50	1546	PPI	-0.0013	0.0018	0.0018	0.934	1.000
forest_mean	50	1546	PPI++	-0.0064	0.0019	0.0019	0.928	0.969
forest_mean	50	1546	Venn-Abers	-0.0027	0.0020	0.0020	0.888	0.923
forest_mean	300	1296	AIPW	0.0011	0.0002	0.0002	0.984	1.036
forest_mean	300	1296	AIPW-EM	0.0007	0.0002	0.0002	0.982	1.012
forest_mean	300	1296	AutoCal	0.0006	0.0002	0.0002	0.978	1.002
forest_mean	300	1296	Labeled-only	0.0007	0.0003	0.0004	0.968	0.706
forest_mean	300	1296	Imputation	-0.0733	0.0000	0.0054	0.000	1281441200512508097500322725888.000
forest_mean	300	1296	IsoCal	0.0010	0.0002	0.0002	0.978	1.000
forest_mean	300	1296	LinearCal	0.0010	0.0002	0.0002	0.984	1.018
forest_mean	300	1296	MonoSpline	0.0010	0.0002	0.0002	0.974	1.004
forest_mean	300	1296	Platt	0.0010	0.0002	0.0002	0.982	1.020
forest_mean	300	1296	PPI	0.0011	0.0002	0.0002	0.986	1.000
forest_mean	300	1296	PPI++	-0.0002	0.0002	0.0002	0.980	0.991
forest_mean	300	1296	Venn-Abers	0.0011	0.0002	0.0002	0.978	1.015
forest_mean	500	1096	AIPW	0.0001	0.0001	0.0001	0.980	1.172
forest_mean	500	1096	AIPW-EM	-0.0002	0.0001	0.0001	0.980	1.188
forest_mean	500	1096	AutoCal	-0.0003	0.0001	0.0001	0.986	1.180
forest_mean	500	1096	Labeled-only	0.0001	0.0002	0.0002	0.972	0.879
forest_mean	500	1096	Imputation	-0.0733	0.0000	0.0054	0.000	7819091278107866750853379522 56.000
forest_mean	500	1096	IsoCal	-0.0000	0.0001	0.0001	0.986	1.215
forest_mean	500	1096	LinearCal	0.0001	0.0001	0.0001	0.982	1.191
forest_mean	500	1096	MonoSpline	0.0000	0.0001	0.0001	0.984	1.207
forest_mean	500	1096	Platt	0.0001	0.0001	0.0001	0.984	1.203
forest_mean	500	1096	PPI	0.0000	0.0002	0.0002	0.982	1.000
forest_mean	500	1096	PPI++	-0.0006	0.0001	0.0001	0.978	1.179
forest_mean	500	1096	Venn-Abers	-0.0001	0.0001	0.0001	0.978	1.206
galaxies_mean	50	16693	AIPW	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	AIPW-EM	0.0035	0.0019	0.0020	0.864	0.944
galaxies_mean	50	16693	AutoCal	0.0019	0.0018	0.0018	0.896	0.997
galaxies_mean	50	16693	Labeled-only	0.0014	0.0041	0.0041	0.890	0.451
galaxies_mean	50	16693	Imputation	-0.0433	0.0000	0.0019	0.000	2376289106442477113652488962048.000
galaxies_mean	50	16693	IsoCal	-0.0011	0.0020	0.0020	0.850	0.908
galaxies_mean	50	16693	LinearCal	0.0030	0.0019	0.0019	0.862	0.973
galaxies_mean	50	16693	MonoSpline	0.0026	0.0020	0.0020	0.830	0.917
galaxies_mean	50	16693	Platt	0.0033	0.0019	0.0019	0.856	0.966
galaxies_mean	50	16693	PPI	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	PPI++	-0.0020	0.0020	0.0020	0.864	0.914
galaxies_mean	50	16693	Venn-Abers	0.0022	0.0020	0.0020	0.830	0.907
galaxies_mean	577	16166	AIPW	-0.0006	0.0001	0.0001	0.906	1.004
galaxies_mean	577	16166	AIPW-EM	-0.0007	0.0001	0.0001	0.906	1.005
galaxies_mean	577	16166	AutoCal	-0.0008	0.0001	0.0001	0.918	0.989
galaxies_mean	577	16166	Labeled-only	-0.0011	0.0003	0.0003	0.910	0.436
galaxies_mean	577	16166	Imputation	-0.0433	0.0000	0.0019	0.000	1873962492481163727822158561 28.000
galaxies_mean	577	16166	IsoCal	-0.0007	0.0001	0.0001	0.894	0.986
galaxies_mean	577	16166	LinearCal	-0.0007	0.0001	0.0001	0.908	1.005
galaxies_mean	577	16166	MonoSpline	-0.0006	0.0001	0.0001	0.892	0.988
galaxies_mean	577	16166	Platt	-0.0007	0.0001	0.0001	0.896	1.002
galaxies_mean	577	16166	PPI	-0.0006	0.0001	0.0001	0.900	1.000
galaxies_mean	577	16166	PPI++	-0.0012	0.0001	0.0001	0.902	0.988
galaxies_mean	577	16166	Venn-Abers	-0.0006	0.0001	0.0001	0.892	0.986
galaxies_mean	1000	15743	AIPW	0.0002	0.0001	0.0001	0.948	0.987
galaxies_mean	1000	15743	AIPW-EM	0.0002	0.0001	0.0001	0.948	0.980
galaxies_mean	1000	15743	AutoCal	0.0001	0.0001	0.0001	0.944	0.977
galaxies_mean	1000	15743	Labeled-only	0.0003	0.0002	0.0002	0.884	0.357
galaxies_mean	1000	15743	Imputation	-0.0433	0.0000	0.0019	0.000	914665280834151199967019008 00.000
galaxies_mean	1000	15743	IsoCal	0.0001	0.0001	0.0001	0.936	0.951
galaxies_mean	1000	15743	LinearCal	0.0002	0.0001	0.0001	0.946	0.980
galaxies_mean	1000	15743	MonoSpline	0.0002	0.0001	0.0001	0.944	0.969
galaxies_mean	1000	15743	Platt	0.0002	0.0001	0.0001	0.942	0.979
galaxies_mean	1000	15743	PPI	0.0002	0.0001	0.0001	0.948	1.000
galaxies_mean	1000	15743	PPI++	-0.0002	0.0001	0.0001	0.946	0.977
galaxies_mean	1000	15743	Venn-Abers	0.0002	0.0001	0.0001	0.942	0.952

Table 2: Representative numerical summary at the smallest, middle, and largest labeled sample sizes for each reproduced benchmark. The reported quantities are Monte Carlo bias, empirical variance of the point estimator, MSE, Wald-interval coverage, and relative efficiency versus PPI for the displayed benchmark set, including PPI++, AIPW-EM, `AutoCal`, `MonoSpline`, and `IsoCal`.